

Research article

Assessing cybersecurity risks and prioritizing top strategies In Vietnam's finance and banking system using strategic decision-making models-based neutrosophic sets and Z number

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ABSTRACT

The financial and banking system in Vietnam is currently confronted with a myriad of cybersecurity risks, a matter of paramount concern for the government, businesses, and individuals alike. These risks threaten the financial system's security, reliability, and integrity, potentially resulting in significant economic losses. This research endeavors to identify the prevailing cybersecurity risks in Vietnam's financial and banking system, assess their impact, and elucidate the interrelationships among these risks. Employing the Multi-Criteria Decision Making (MCDM) approach, the study integrates the DELPHI technique, Decision-Making Trial and Evaluation Laboratory (DEMATEL), and Combined Compromise Solution (COCOSO) methods, complemented by Neutrosophic Sets (NS) and Z-number concepts to enhance the accuracy and reliability of the findings. The research findings reveal the existence of 15 cybersecurity risks in Vietnam's financial and banking system, with Malware Infections and Supply Chain Vulnerabilities emerging as the most consequential risks. Moreover, the study identifies investing in advanced threat detection systems as the optimal strategy for mitigating cybersecurity risks in Vietnam. The results underscore the importance of addressing these critical risks to safeguard the financial infrastructure, focusing on deploying robust cybersecurity measures to enhance overall system resilience.

1. Introduction

Cybersecurity in the finance and banking sector, particularly in Vietnam, is paramount as we transition to a digital economy [1]. The financial industry is a prime target for cybercriminals due to its high volume of valuable financial data and assets, resulting in escalating cybersecurity risks. In Vietnam, the digital transformation of the banking system is rapidly underway, with most banks implementing or developing their digital strategies in response to pervasive technological advancements [2]. Despite the recognized importance of cybersecurity, Vietnam's finance and banking sector faces several challenges in effectively addressing cyber risks. These challenges include the constantly evolving nature of cyber threats, the complexity of financial systems and technologies, the shortage of skilled cybersecurity professionals, and the lack of a comprehensive risk management framework [3]. Moreover, the globalization of

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financial services and the interconnectedness of institutions have expanded the potential attack surface, complicating the identification and mitigation of cyber risks. Moreover, cybersecurity enables banks to prevent and respond to hacks, unauthorized access, data breaches, and other online threats [4].

Cybersecurity is critical to the financial industry's success, protecting sensitive customer data, ensuring the integrity of financial transactions, and confirming compliance with regulatory requirements [5]. Robust cybersecurity strategies also help banks secure funds and prevent financial losses. Additionally, cybersecurity allows banks to prevent and react to hacks, unauthorized access, data breaches, and other online threats. Cybersecurity risks encompass threats to the confidentiality, integrity, and availability of information and systems, posing potential adverse impacts on organizational operations, assets, and national security [6,7]. Financial institutions' escalating digitization and interconnectivity heighten vulnerability to evolving threats, jeopardizing sensitive data, operations, and stakeholder trust. Beyond financial losses, cybersecurity breaches entail reputational damage, loss of customer trust, and regulatory repercussions [8].

Vietnam's emergence as a regional financial hub intensifies the urgency of addressing these risks, underscoring the imperative to comprehend, mitigate, and manage cybersecurity threats to safeguard the stability and integrity of the financial and banking system in today's digitally interconnected landscape [9]. Given the complexities and uncertainties associated with cybersecurity risks, traditional risk assessment methods may fall short of providing a holistic and practical approach to decision-making [10,11]. There is a pressing need for a comprehensive strategic decision-making framework that integrates expert opinions, considers the interrelationships between various risk factors and mitigation strategies, and accounts for the inherent uncertainty and vagueness in cybersecurity assessments [12]. Therefore, this study aims to formulate comprehensive strategic decision-making models tailored to the dynamic and challenging business environment characterized by Turbulence, Uncertainty, Novelty, and Ambiguity (TUNA) using the MCDM methods. The research objectives are as follows.

- (i) Identify and assess the critical cybersecurity risks in Vietnam's finance and banking system.
- (ii) Analyze the potential impact and the interconnectedness of crucial cybersecurity risks in Vietnam's finance and banking system.
- (iii) Prioritize strategies for mitigating these risks using strategic decision-making models, incorporating uncertainty and vagueness using Neutrosophic sets and Z numbers.

While cybersecurity risk analysis is addressed in academic literature, few studies approach it as a multi-criteria decision-making problem [13]. Moreover, empirical examinations of cybersecurity risks and mitigation strategies in Vietnam's finance and banking sector using MCDM models are lacking. Hence, this research aims to bridge this gap by addressing vital questions.

- (i) What are key cybersecurity risks in Vietnam's finance and banking system?
- (ii) How do these key cybersecurity risks correlate in Vietnam's finance and banking system?
- (iii) How can the proposed model be validated for prioritizing strategies for mitigating the identified cybersecurity risks?

In this study, Neutrosophic sets (NS) are a significant advancement in fuzzy set theory [14,15]. Smarandache [16] proposed that they present uncertain, incomplete, imprecise, and indeterminate information in real-world problems. NS is a generalization of the fuzzy set. It combines the concepts of fuzzy sets, and neutrosophic sets, where fuzzy sets are used to tackle uncertainty using the membership grade, and neutrosophic sets are used to tackle uncertainty using the truth, indeterminacy, and falsity membership grades, which are considered independent [17]. This study gains access to more comprehensive and accurate information about experts' responses by leveraging neutrosophic sets. Consequently, the calculation results are expected to more faithfully reflect reality, enhancing the overall quality and reliability of the research findings. In addition to the general form of NS, the Single-Valued Neutrosophic Set (SVNS) is proposed as a specific instance that is particularly useful for real-world scientific and engineering applications [18–21]. This assumption is beneficial in various scenarios, such as information fusion, where data from different sensors must be integrated. SVNS, being a subset of NS, utilizes single-valued memberships, thereby inheriting the mathematical properties of NS [18–20]. In this study, we employ the concept of NS to establish a general concept and build upon previous research that combines neutrosophic fuzzy sets with Z-numbers. In decision-making, assessments often occur amidst incomplete information and uncertainty [22]. Compared to other versions of fuzzy sets such as Fuzzy Sets [23], Hexagon fuzzy numbers [24,25], Type-2 fuzzy sets [26], Intuitionistic fuzzy sets (IFSs) [27], Hesitant fuzzy sets (HFSs) [28], Pythagorean fuzzy sets (PyFSs) [29], Picture fuzzy sets (PFSs) [30], Spherical fuzzy sets (SFSs) [31,32], T- Spherical fuzzy sets (T-SFSs) [33], NS have unique characteristics as below.

Unlike ordinary fuzzy sets [23], which consider only the degree of membership, NS encompasses an object's degree of truth, indeterminacy, and falsity. This comprehensive representation enables a more accurate modeling of real-world situations. For instance, in medical diagnosis, patient symptoms may not definitively indicate a specific disease (truth) nor entirely rule it out (falsity) while also exhibiting some ambiguity (indeterminacy). NS can capture this complexity more effectively than fuzzy sets. Additionally, while Type-2 fuzzy sets were developed to address uncertainties, they lack the capability to handle indeterminacy. By incorporating an indeterminacy membership function, NS can effectively manage such situations. For example, in weather forecasting, predictions may be uncertain (Type-2 fuzzy set), but indeterminacy arising from climate change can be modeled using NS. Assuming $\alpha(x)$, $\beta(x)$, and $\gamma(x)$ represent memberships for truth, indeterminacy, and falsity, respectively, with $x \rightarrow [0, 1]$, we have $0 \leq \alpha(x) + \beta(x) + \gamma(x) \leq 3$. Unlike other fuzzy types, the sum of the three memberships in the NS can reach a maximum value of 3, where the maximum sum is typically limited to 1. This characteristic allows for a broader observation range, enhancing accuracy, which distinguishes NS from other fuzzy types.

IFSs, introduced by Atanassov [27] in 1986, consider an object's degree of membership and non-membership. Although IFSs have

notable advantages, they still face limitations when the sum of membership degree (MD) and non-membership degree (NMD) exceeds 1 [34,35]. For instance, assume $\omega(x)$ represents MD and $\vartheta(x)$ represents NMD of an IFSs number; the constraint is typically defined as $0 \leq \omega(x) + \vartheta(x) \leq 1$. However, when $\omega(x) = 0.6$ and $\vartheta(x) = 0.7$, the sum of MD and NMD equals 1.3, surpassing the limit of 1 set by IFSs. In this case, NS, with a maximum number of memberships equal to three, offers a solution to this issue. Furthermore, IFSs do not account for the degree of indeterminacy. NS, conversely, considers an object's degree of truth, indeterminacy, and falsity, thus providing a more comprehensive representation. For example, in predicting stock market trends, the prediction might be partially true (bullish trend), partially false (bearish trend), and partially indeterminate (market volatility). Neutrosophic sets can model this situation more accurately than IFSs.

HFSs, introduced by Torra [28] in 2010, allow decision-makers to assign multiple possible membership degrees to an element, offering flexibility but lacking in handling indeterminacy. NS, with its indeterminacy membership function, addresses this limitation. For instance, in group decision-making, differing expert opinions and consensus ambiguity can be captured using Neutrosophic sets. Additionally, while the maximum membership value of HFSs does not exceed 1, the total membership of NS can reach up to 3, underscoring the superiority of NS over HFSs.

PyFSs, introduced by Yager [29] in 2013, extend the concept of IFSs by allowing the sum of the membership and non-membership degrees to be greater than 1 in many cases. In some instances, even with PyFSs, the sum of the squared MD and NMD values may still exceed 1, violating the condition. Assuming ω represents MD and ϑ represents NMD of a PyFSs, the constraint is typically defined as $0 \leq \omega(x)^2 + \vartheta(x)^2 \leq 1$. With $\omega(x) = 0.7$ and $\vartheta(x) = 0.8$, $\omega(x)^2 + \vartheta(x)^2$ results in 1.13, contravening the PyFSs condition. Therefore, NS, which allows for a maximum total membership of 3, offers an advantageous solution in such scenarios. Similar to IFSs, PyFSs also still do not account for the degree of indeterminacy. NS can accommodate truth, indeterminacy, and falsity and offer a more precise modeling approach than IFSs, HFSs, and PyFSs.

PFSs, proposed by Cuong and Kreinovich [30] in 2014, introduce a new membership degree called "hesitation" to handle uncertainty. While this provides a certain level of flexibility, it cannot still comprehensively handle indeterminacy. For instance, in a weather prediction scenario, the forecast might be partially sunny (membership), partially not sunny (non-membership), and partially uncertain due to unpredictable factors (indeterminacy). With their indeterminacy membership function, Neutrosophic sets can handle this better than PFS. Furthermore, similar to IFS, PyFS, or HFS, the constraint that the total membership of PFS does not exceed one may be violated in certain cases [36]. For instance, considering $\mu(x)$ as the membership grades, $\nu(x)$ as the neutrality level, and $\lambda(x)$ as the non-membership level of a number in PFS, the condition is $0 \leq \mu(x) + \nu(x) + \lambda(x) \leq 1$. With $\mu(x) = 0.5$, $\nu(x) = 0.4$, and $\lambda(x) = 0.3$, the sum $\mu(x) + \nu(x) + \lambda(x)$ equal to 1.2, which contravenes the condition. In contrast, NS allows for a total of three memberships, equating to 1.2, demonstrating the advantages of NS over PFS.

To address situations where the sum of three PFSs memberships exceeds one, SFSs, introduced by Gündoğdu and Kahraman [32] in 2019, have been proposed with the primary concept being the square of three PFS memberships [22,37–39]. Assuming $\mu(x)$ represents the membership degree, $\nu(x)$ the membership degree, and $\lambda(x)$ the membership degree of a number in PFSs, the condition is $0 \leq \mu(x)^2 + \nu(x)^2 + \lambda(x)^2 \leq 1$. Squaring can resolve many instances where the sum of three memberships exceeds 1; for instance, the previous case where $\mu(x)^2 + \nu(x)^2 + \lambda(x)^2 = 0.5$, no longer violates the assumption. However, if we consider $\mu(x) = 0.8$, $\nu(x) = 0.7$, and $\lambda(x) = 0.6$, the value of $\mu(x)^2 + \nu(x)^2 + \lambda(x)^2$ equals 1.49, thus contravening the assumption of SFSs. Nevertheless, NS accommodates cases where the number of memberships exceeds one, underscoring its advantages over PFSs and SFSs.

Considering membership's observational meaning, NS also offers unique advantages over SFSs and T-Spherical Fuzzy Sets (T-SFSs). SFSs and T-SFSs, proposed by Ullah et al. [33] in 2020, are extensions of fuzzy sets that incorporate the concept of direction in decision-making. However, neither SFS nor T-SFS explicitly account for the degree of indeterminacy. For example, in a decision-making scenario involving investment options, an investor might be partially inclined towards an option (membership), partially disinclined (non-membership), and partially uncertain due to market volatility (indeterminacy). While SFSs and T-SFSs can capture the investor's inclination towards or away from an option, they do not adequately handle the uncertainty aspect. With their ability to handle truth, indeterminacy, and falsity, NS can model this situation more accurately. Moreover, NS provides a more flexible and comprehensive tool for dealing with such uncertainties in complex decision-making scenarios where the data is incomplete, inconsistent, or uncertain. Thus, despite the advancements brought by SFSs and T-SFSs, NS still holds a significant edge when it comes to handling uncertainty, imprecision, and indeterminacy in real-world situations.

In decision-making, uncertainties are ubiquitous, and decisions often arise in contexts where information is incomplete [40–42]. Hence, acknowledging the inherent uncertainty in decision-making processes is crucial [43]. Z-numbers, pioneered by Zadeh [44], emerged as indispensable tools in response to this need. Specifically designed for computations in uncertain and incompletely reliable environments, Z-numbers offers a structured approach to handling uncertainty and incomplete information. Z-numbers comprise two integral components: the Z-number and fuzzy information, encapsulating the assessment score and the associated reliability level [44]. Thus, Z-numbers offer a significant complement to NS. While NS excels in handling uncertainty, imprecision, and indeterminacy, they lack explicit consideration of the reliability of the information source. This is where Z-numbers play a crucial role. As an extension of fuzzy numbers, Z-numbers incorporate a measure of reliability [45]. For instance, imagine two financial analysts offering company growth forecasts. While NS can represent the analysts' forecasts in terms of truth, indeterminacy, and falsity, they overlook the analysts' reliability. Conversely, Z-numbers can model both the forecast (the "restriction") and the reliability of the analyst (the "reliability"). This enables a more thorough analysis, as decision-makers can now weigh both the forecast and the source's reliability. Consequently, Z-numbers complement Neutrosophic sets, enhancing decision-making in scenarios where the source's reliability is critical.

The motivation and novelty of this research stem from the proposal and application of NS combined with the Z-number concept in

MCDM methods to assess cybersecurity risks and prioritize top strategies in Vietnam's finance and banking system. Integrating Z-numbers and NS (NZN) is essential to fully leveraging their strengths. This integration establishes a robust system capable of handling uncertainty, imprecision, and indeterminacy while considering the reliability of the information source. Such an approach enhances the accuracy and reliability of decision-making processes across various domains, from finance to engineering. Initially gathered in linguistic form, expert responses are converted to NZN for computation using the MCDM method. This combination allows for a comprehensive and nuanced analysis of uncertain and incomplete information, improving the quality and reliability of decision-making.

This study proposes a three-stage MCDM method of DELPHI, DEMATEL, and COCOSO in conjunction with NZN. The DELPHI method validates the importance and relevance of identified potential risks. Subsequently, risks validated through the NZN DELPHI phase undergo weighting for influence and analysis of cause-and-effect relationships using the NZN DEMATEL method. Finally, proposed strategies are ranked for effectiveness and utility through the NZN COCOSO method. Additionally, a comparative analysis is conducted using results obtained from the NZN Technique for Order Performance By Similarity To Ideal Solution (TOPSIS) and NZN Multiplicative Form Of Multiple Objectives (MULTIMOORA) methods. Notably, including a variable validation step using NZN DELPHI enhances the thoroughness of risk filtering and validation, thereby elevating the overall quality of the research compared to prior studies.

As a result, this study makes several significant contributions to cybersecurity risk management within Vietnam's finance and banking sectors. Firstly, it proposes and develops comprehensive strategic decision-making models tailored to the dynamic and challenging business environment characterized by TUNA. By integrating methods with NZN, the study offers novel approaches for assessing, prioritizing, and mitigating cybersecurity risks. Secondly, through empirical analysis and expert consultations, the study identifies and assesses vital cybersecurity risks specific to Vietnam's finance and banking system, contributing to a deeper understanding of the threat landscape and enabling stakeholders to address vulnerabilities proactively. In terms of theoretical and mathematical advancements, this study distinguishes itself from previous research utilizing the same methodology by introducing a novel approach for calculating expert weights based on their experience and education level. It also proposes a pooling formula incorporating these expert weights, as detailed in Section 3. By considering expert weights, the study enhances the accuracy and reliability of the results, acknowledging that experts possess varying qualifications and experience. This methodological enhancement ensures that the assessments reflect a more nuanced and credible aggregation of expert opinions, thereby contributing significantly to the robustness of the study's findings. Thirdly, by employing the DEMATEL method, causal relationships between risks are elucidated, providing insights into their influence and interrelationships, thus aiding in the analysis of interconnectedness and potential impact of identified cybersecurity risks within the finance and banking system.

Furthermore, the study introduces a model and associated calculation formulas for the NZN DELPHI method, thereby validating the variables based on expert opinions. This methodological contribution enhances the rigor of the validation process and ensures that the variables are assessed with greater precision and reliability. By leveraging expert insights through the NZN DELPHI method, the study

Table 1
Related Work applied NS and Z Number.

No	Authors	Methods	Characteristics					
			Uncertainty	Expert weight	Criteria weight	Relations	Ranking	Reliability
1	Mohamad Sharaf et al. [46]	Type-2 Neutrosophic Fuzzy-Weighted Zero-Inconsistency and Type-2 Neutrosophic Fuzzy Decision by Opinion Score Method	✓		✓		✓	
2	Lu et al. [47]	Simplified neutrosophic TOPSIS	✓				✓	
3	Ye et al. [48]	Aczel–Alsina Weighted Aggregation Operators of Neutrosophic Z-Numbers	✓		✓			✓
4	Diznarda et al. [14]	Neutrosophic AHP and DEMATEL	✓		✓	✓		
5	Shao et al. [49]	Z-DEMATEL and Z-TOPSIS	✓		✓	✓	✓	✓
6	Eldrandaly et al. [50]	Neutrosophic AHP and DEMATEL	✓		✓	✓		
7	Yong et al. [51]	Trapezoidal neutrosophic Z-numbers	✓				✓	✓
8	Du et al. [52]	NZN weighted arithmetic averaging and NZN weighted geometric averaging operators	✓				✓	✓
9	Yörükoğlu and Aydın [53]	Neutrosophic TOPSIS	✓				✓	
10	Peng and Smarandache [54]	Neutrosophic CRITIC and COCOSO	✓		✓		✓	
11	Haktanir and Kahraman [55]	Z-fuzzy numbers	✓					✓
12	Nabeeh et al. [56]	Neutrosophic AHP	✓		✓			
13	Ye [57]	Neutrosophic number linear programming	✓					
14	Zavadskas et al. [58]	Neutrosophic MULTIMOORA	✓		✓			
	This Study	Fully completed NZN DELPHI-DEMATEL-COCOSO	✓	✓	✓	✓	✓	✓

provides a robust framework for evaluating complex issues, thereby significantly advancing the field's understanding and application of expert-driven validation techniques. Additionally, using the COCOSO algorithm, the study prioritizes mitigation strategies for addressing identified cybersecurity risks, incorporating uncertainty and vagueness through neutrosophic sets and Z numbers, thereby offering robust and reliable decision-making frameworks for stakeholders to implement effective risk mitigation measures. Finally, through empirical analysis and comparative assessments, the study validates the proposed models for prioritizing mitigation strategies, enhancing confidence in their utility and reliability. Overall, these contributions are expected to have significant implications for industry stakeholders, policymakers, and researchers, facilitating proactive risk mitigation efforts and strengthening the resilience of financial systems in the face of evolving cyber threats.

The remainder of this research consists of four sections. Section 2 will review previous studies and present the research model. Section 3 will elucidate the concepts and calculation formulas associated with each method. Section 4 presents the case study and analyzes the results. Finally, section 5 will summarize and conclude the research findings.

2. Literature review

2.1. Literature review on established method

Recent years have witnessed a surge in research efforts to leverage advanced mathematical models to address complex decision-making problems. In particular, NZN has emerged as a promising approach to tackling uncertainties in various real-world scenarios. Table 1 provides an overview of related works that have applied NS and Z fuzzy numbers in decision-making processes.

In the literature review, Zavadskas et al. [58] proposed Neutrosophic MULTIMOORA to select construction materials and elements for single-family house construction, offering a holistic approach that fills a significant gap in residential construction. Conversely, Ye [57] employed Neutrosophic Number Linear Programming (NNLP) to optimize resource allocation in complex systems, showcasing its potential to enhance decision-making processes through practical applications in production planning. Additionally, Nabeeh et al. [56] introduced a Neutrosophic AHP approach for IoT enterprises, illustrated through case studies in various locations. Peng et al. [54] prioritized dimensions of rare earth industry security using CRITIC and Neutrosophic Soft Decision Making, addressing complex security challenges. Yörükoğlu et al. [53] utilized Neutrosophic TOPSIS in intelligent container evaluation, emphasizing its role in Industry 4.0 Supply Chain Management Systems.

Furthermore, Diznarda et al. [14] and Eldrandaly et al. [50] applied Neutrosophic AHP and DEMATEL methodologies to analyze decision-making scenarios in tourism and construction project management, respectively. Lu et al. [47] introduced Simplified Neutrosophic TOPSIS for evaluating teaching quality in Chinese colleges, highlighting its importance in enhancing teaching abilities. Lastly, Mohamad Sharaf et al. [46] addressed challenges in evaluating 5G-RANs architecture with Type-2 Neutrosophic Fuzzy-Weighted Zero-Inconsistency and Type-2 Neutrosophic Fuzzy Decision methods. Despite these advancements, none have comprehensively addressed the issue of data reliability and certainty derived from expert responses, nor have they adequately considered the weighting of experts based on their qualifications and experience. This highlights a potential area for further research and exploration in the field.

Integrating Z-Number concepts with neutrosophic fuzzy sets to form NZN presents a promising solution to bridge existing gaps in research methods. For instance, Du et al. [52] introduced NZN, which combines truth, falsity, and indeterminacy degrees with reliability degrees, offering a comprehensive framework for decision-making. They further developed NZN-weighted arithmetic (NZNWA) and NZN-weighted geometric averaging (NZNWGA) operators to aggregate NZN information, enhancing decision-making processes. In a similar vein, Yong et al. [51] proposed trapezoidal neutrosophic Z-numbers (TrNZNs) to overcome limitations in existing trapezoidal neutrosophic numbers (TrNNs) and NZNs, introducing TrNZN weighted arithmetic averaging (TrNZNWA) and TrNZN weighted geometric averaging (TrNZNWGA) operators for improved MCDM methods. Additionally, Haktanir et al. [55] demonstrated the applicability of Z-fuzzy hypothesis testing in both left and right-sided scenarios, along with sensitivity analyses, highlighting the effectiveness of their approach in handling uncertain and imprecise data within hypothesis testing contexts. Through these advancements, integrating Z-number concepts with neutrosophic fuzzy sets offers a robust framework for addressing uncertainties and enhancing decision-making processes in various domains.

The fusion of neutrosophic fuzzy sets and Z numbers has recently represented a recent advancement, offering a more nuanced and comprehensive approach to decision-making and sustainability assessment—the research conducted by Ye et al. [48,59] significantly contribute to the field by introducing specialized operations and weighted aggregation methods designed for NZN. Their study enhances the adaptability of decision-making processes by allowing for parameter adjustments that align with decision-makers' preferences. Specifically, they introduce Aczel–Alsina operations tailored for NZNs, including Aczel–Alsina t-norm and t-conorm operations, as well as NZN Aczel–Alsina weighted arithmetic averaging (NZNAAWAA) and NZN Aczel–Alsina weighted geometric averaging (NZNAAWGA) operators for effectively aggregating NZNs. In a parallel effort, Shao et al. [60] present a holistic framework for evaluating smart cities' sustainable development (SD) level. Their approach involves utilizing the Z-DEMATEL technique to identify mutual influence relationships and derive influence weights. Additionally, they employ the Z-TOPSIS-AL approach to assess sustainable development performance comprehensively. These contributions underscore the potential of integrating neutrosophic fuzzy sets and Z numbers to enhance decision-making processes and sustainability assessments in complex systems.

2.2. Literature review on risk factor and solution strategy

Many prior studies have extensively examined cybersecurity risks within the financial and banking sector. Shulha et al. [8] utilized

the fuzzy cognitive map method to identify and analyze 13 cybersecurity risks impacting banking information resources. Meanwhile, Ahmed et al. [10] employed the Graph Theory and Matrix Approach (GTMA) to assess cybersecurity challenges in the context of Industry 5.0 and determine their relative importance. Additionally, Sarefo et al. [61] conducted a cybersecurity risk study within the national framework of Botswana. These studies have employed diverse methodologies to pinpoint and analyze cybersecurity risks affecting banking systems and businesses, often within specific national contexts.

To commence this analysis, a comprehensive search was conducted across prominent library databases, including Science Direct, Scopus, and Web of Science. Various keywords and search strings such as "cybersecurity risk," "Vietnam's finance and banking system," "finance system cybersecurity risk," and "banking system cybersecurity risk" were employed to retrieve relevant scholarly papers. As a result of this search, the study identified 17 potential cybersecurity risks impacting Vietnam's finance and banking system. These 17 factors are presented in Table 2.

Ahmed et al. [10] highlighted the risks associated with cloud security, supply chain, and emerging technologies risks. Javaheri et al. [13], in their comprehensive study on cybersecurity threats in FinTech, identified well-known types of cyberattacks, including Advanced Persistent Threats (APTs), Denial-of-Service (DoS) Attacks, Ransomware Attacks, and Malware Infections. Additionally, his study pointed out risks stemming from poor system architecture, such as Data Breaches, Weak Authentication Systems, and Internet of Things (IoT) Vulnerabilities. Human-related risks, such as Insider Threats and Regulatory Compliance Risks, were also mentioned. Shulha et al. [8], In their study on Banking Information Resource Cybersecurity System Modeling, they emphasized the risk of man-in-the-middle attacks, highlighting the human factor in cybersecurity. This research also discussed system security risks, including Mobile Device and Web Application Security Risks, as well as Phishing Attacks. Malware Infections were again noted as a significant threat. Thus, previous studies on cybersecurity in the banking sector have addressed a wide range of risks, including various forms of cyberattacks, vulnerabilities in system architecture, and human factors. This study builds on the findings of these prior studies, applying them to the Vietnamese context to rank and identify root cause factors for focused resolution.

Drawing upon the insights from the literature review, the research team presents ten potential strategies to mitigate and address the identified risks. These strategies, supported by findings from previous studies in related fields, are summarized in Table 3.

3. Methodology

3.1. Research process

Building upon this body of research, this study proposes a comprehensive model to investigate cybersecurity risks in Vietnam's financial and banking system using the MCDM approach, which encompasses the DELPHI, DEMATEL, and COCOSO methods combined with NZN. The research model is illustrated in Fig. 1:

The study comprises three distinct phases. Firstly, potential cybersecurity risks are identified through an extensive literature review process. Subsequently, these risks undergo validation for relevance and appropriateness utilizing the DELPHI method combined with NZN. This involves synthesizing expert opinions gathered via questionnaires, where experts assess the importance and relevance of the identified risks. Notably, the study assigns weights to experts based on their years of experience and educational qualifications. This information is converted into NZN numbers, aggregated, and crisp values. The weights assigned to experts are utilized throughout all research phases and multiplied by each expert's corresponding opinion. Following the compilation of opinions from 30 experts, a threshold value is calculated. Risks with assessment values falling below the threshold value are subsequently rejected from further consideration.

In phase 2, experts will assess the level of mutual influence among the accepted risks identified in phase 1. This evaluation uses the DEMATEL method combined with the NZN concept. The DEMATEL technique, pioneered by Fontela and Gabus in 1976 [74], has

Table 2
List of cybersecurity risks potentially affecting Vietnam's finance and banking system.

Risks	References
TH1	Cloud Security Risks [10]
TH2	Advanced Persistent Threats (APTs) [13,61]
TH3	Mobile Device and Web Application Security Risks [8]
TH4	Supply Chain Vulnerabilities [10]
TH5	Denial-of-Service (DoS) Attacks [8,13]
TH6	Ransomware Attacks [13,61]
TH7	Data Breaches [8,13]
TH8	Phishing Attacks [8,61]
TH9	Insider Threats [13]
TH10	Biometric Data Vulnerabilities [62]
TH11	Weak Authentication Systems [13]
TH12	Man-in-the-Middle Attacks [8]
TH13	Regulatory Compliance Risks [13]
TH14	Malware Infections [8,13,61]
TH15	Emerging Technologies Risks [10]
TH16	Internet of Things (IoT) Vulnerabilities [10,13]
TH17	Insecure Application Programming Interfaces (APIs) [63]

Table 3
List of potential strategies.

Strategies	References
SG1	Establish a Comprehensive Incident Response Plan [64]
SG2	Collaborate with Third-Party Vendors [65]
SG3	Implement Robust Employee Training Programs [66]
SG4	Conduct Regular Security Audits and Penetration Testing [67]
SG5	Implement Encryption Protocols [68]
SG6	Adopt a Zero-Trust Security Model [69]
SG7	Enhance Endpoint Security [70]
SG8	Regularly Update and Patch Systems [71]
SG9	Enforce Multi-Factor Authentication (MFA) [72]
SG10	Invest in Advanced Threat Detection Systems [73]

effectively addressed complex global issues across scientific, political, and economic domains, leveraging expert judgments. This method yields insights into the weight of influence attributed to each risk and delineates cause-and-effect relationships among them. Notably, to preserve information integrity, this study employs defuzzification after calculating the total influence matrix, ensuring all matrix calculations adhere to the NZN format. In phase 3, experts will assess strategies based on their efficacy in mitigating risks. The data will be analyzed using the COCOSO method in conjunction with NZN, generating rankings for these strategies. Notably, the ranking results will undergo sensitivity and comparative analysis, juxtaposing them with rankings derived from other calculation methods, notably TOPSIS and MULTIMOORA, which incorporate the NZN number concept. This comprehensive evaluation approach aims to ensure the robustness and reliability of the ranking outcomes. The study utilized Microsoft Excel 2016 and SPSS version 26 to analyze the data. Processing the responses from 30 experts, encompassing 17 factors and 10 solutions, required an average of 180 min per analysis.

3.2. Neutrosophic Z-number sets

Z-number, as proposed by Zadeh [44], comprises an ordered pair of fuzzy numbers (A, C) associated with an uncertain variable U, representing the fuzzy value of U (A) and a reliability measure (C) of A. However, this representation lacks information regarding the indeterminacy and falsity aspects of Z-numbers. To address this limitation and encompass comprehensive information about the truth, indeterminacy, and falsity of Z-numbers, the concept of a NZN can be introduced as an extension of the Z-number framework [52].

Definition 1 [52] Let X denote a universe set. A NZN set in X is defined using Equation (1):

$$N_z = \{[x, \alpha(A, C)(x), \beta(A, C)(x), \gamma(A, C)(x)] | x \in X\} \quad (1)$$

Where $\alpha(A, C)(x) = (\alpha_A(x), \alpha_C(x))$; $\beta(A, C)(x) = (\beta_A(x), \beta_C(x))$; $\gamma(A, C)(x) = (\gamma_A(x), \gamma_C(x)) : X \rightarrow [0, 1]$ are the order pairs of truth, indeterminacy, and falsity fuzzy values. A signifies neutrosophic values pertaining to the universe set X, and C signifies neutrosophic measures of reliability associated with A. These components adhere to the specified conditions:

$$0 \leq \alpha_A(x) + \beta_A(x) + \gamma_A(x) \leq 3 \text{ and } 0 \leq \alpha_C(x) + \beta_C(x) + \gamma_C(x) \leq 3$$

For the sake of clarity and convenience, the element $[x, \alpha(A, C)(x), \beta(A, C)(x), \gamma(A, C)(x)]$ in N_z is succinctly represented as $N_z = [\alpha(A, C), \beta(A, C), \gamma(A, C)] = [(\alpha_A, \alpha_C), (\beta_A, \beta_C), (\gamma_A, \gamma_C)]$, name NZN

Definition 2 [52,75] Let $N_{z1} = [\alpha_1(A, C), \beta_1(A, C), \gamma_1(A, C)] = [(\alpha_{A1}, \alpha_{C1}), (\beta_{A1}, \beta_{C1}), (\gamma_{A1}, \gamma_{C1})]$ and $N_{z2} = [\alpha_2(A, C), \beta_2(A, C), \gamma_2(A, C)] = [(\alpha_{A2}, \alpha_{C2}), (\beta_{A2}, \beta_{C2}), (\gamma_{A2}, \gamma_{C2})]$ be two NZNs and $\varepsilon > 0$. Then, we give the following relations using Equations (2)–(10):

$$1. N_{z1} \supseteq N_{z2} \iff \alpha_{A1} \geq \alpha_{A2}, \alpha_{C1} \geq \alpha_{C2}, \beta_{A1} \leq \beta_{A2}, \beta_{C1} \leq \beta_{C2}, \gamma_{A1} \leq \gamma_{A2} \text{ and } \gamma_{C1} \leq \gamma_{C2} \quad (2)$$

$$2. N_{z1} = N_{z2} \iff N_{z1} \supseteq N_{z2} \text{ and } N_{z2} \supseteq N_{z1} \quad (3)$$

$$3. N_{z1} \cup N_{z2} \iff [(\alpha_{A1} \vee \alpha_{A2}, \alpha_{C1} \vee \alpha_{C2}), (\beta_{A1} \wedge \beta_{A2}, \beta_{C1} \wedge \beta_{C2}), (\gamma_{A1} \wedge \gamma_{A2}, \gamma_{C1} \wedge \gamma_{C2})] \quad (4)$$

$$3. N_{z1} \cap N_{z2} \iff [(\alpha_{A1} \wedge \alpha_{A2}, \alpha_{C1} \wedge \alpha_{C2}), (\beta_{A1} \vee \beta_{A2}, \beta_{C1} \vee \beta_{C2}), (\gamma_{A1} \vee \gamma_{A2}, \gamma_{C1} \vee \gamma_{C2})] \quad (5)$$

$$4. (N_{z1})^c = [(\gamma_{A1}, \gamma_{C1}), (1 - \beta_{A1}, 1 - \beta_{C1}), (\alpha_{A1}, \alpha_{C1})] \text{ (Complement of } N_{z1}) \quad (6)$$

$$5. N_{z1} \oplus N_{z2} = [(\alpha_{A1} + \alpha_{A2} - \alpha_{A1}\alpha_{A2}, \alpha_{C1} + \alpha_{C2} - \alpha_{C1}\alpha_{C2}), (\beta_{A1}\beta_{A2}, \beta_{C1}\beta_{C2}), (\gamma_{A1}\gamma_{A2}, \gamma_{C1}\gamma_{C2})] \quad (7)$$

$$6. N_{z1} \otimes N_{z2} = [(\alpha_{A1}\alpha_{A2}, \alpha_{C1}\alpha_{C2}), (\beta_{A1} + \beta_{A2} - \beta_{A1}\beta_{A2}, \beta_{C1} + \beta_{C2} - \beta_{C1}\beta_{C2}), (\gamma_{A1} + \gamma_{A2} - \gamma_{A1}\gamma_{A2}, \gamma_{C1} + \gamma_{C2} - \gamma_{C1}\gamma_{C2})] \quad (8)$$

$$7. \varepsilon N_{z1} = [(1 - (1 - \alpha_{A1})^\varepsilon, 1 - (1 - \alpha_{C1})^\varepsilon), (\beta_{A1}^\varepsilon, \beta_{C1}^\varepsilon), (\gamma_{A1}^\varepsilon, \gamma_{C1}^\varepsilon)] \quad (9)$$

$$8. (N_{z1})^\varepsilon = [(\alpha_{A1}^\varepsilon, \alpha_{C1}^\varepsilon), (1 - (1 - \beta_{A1})^\varepsilon, 1 - (1 - \beta_{C1})^\varepsilon), (1 - (1 - \gamma_{A1})^\varepsilon, 1 - (1 - \gamma_{C1})^\varepsilon)] \quad (10)$$

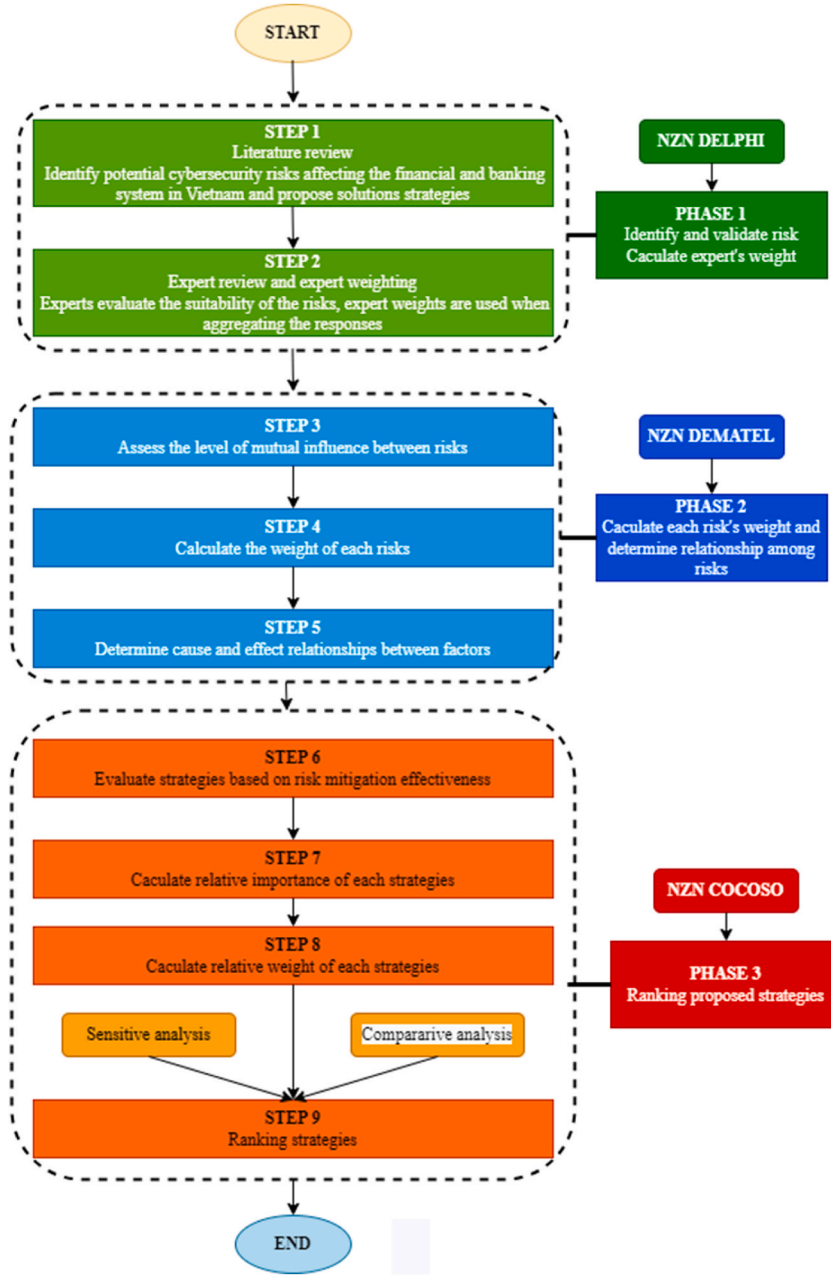


Fig. 1. Research framework.

To defuzzy $N_{z1} = [\alpha_1(A, C), \beta_1(A, C), \gamma_1(A, C)] = [(\alpha_{A1}, \alpha_{C1}), (\beta_{A1}, \beta_{C1}), (\gamma_{A1}, \gamma_{C1})]$, using Equation (11):

$$DEF(N_{z1}) = \frac{2 + \alpha_{A1} \alpha_{C1} - \beta_{A1} \beta_{C1} - \gamma_{A1} \gamma_{C1}}{3} \text{ for } DEF(N_{z1}) \in [0, 1] \quad (11)$$

Illustrative example 1: Let two numbers NZN: $N_{z1} = [(0.8, 0.8), (0.15, 0.15), (0.2, 0.2)]$ and $N_{z2} = [(0.6, 0.8), (0.35, 0.15), (0.4, 0.2)]$, $\varepsilon = 0.7$, an example of Equations (7)–(11) are shown below:

$$N_{z1} \oplus N_{z2} = [(0.8, 0.8), (0.15, 0.15), (0.2, 0.2)] \oplus [(0.6, 0.8), (0.35, 0.15), (0.4, 0.2)] = [(0.92, 0.96), (0.0525, 0.0225), (0.08, 0.04)]$$

$$\begin{aligned} N_{z1} \otimes N_{z2} &= [(0.8, 0.8), (0.15, 0.15), (0.2, 0.2)] \otimes [(0.6, 0.8), (0.35, 0.15), (0.4, 0.2)] \\ &= [(0.48, 0.64), (0.4475, 0.2775), (0.52, 0.36)] \end{aligned}$$

$$\varepsilon N_{Z1} = 0.7 \cdot [(0.8, 0.8), (0.15, 0.15), (0.2, 0.2)] \\ = [(0.6759, 0.6759), (0.265, 0.265), (0.3241, 0.3241)]$$

$$(N_{Z1})^\varepsilon = ([[(0.8, 0.8), (0.15, 0.15), (0.2, 0.2)]])^{0.7} \\ = [(0.8554, 0.8554), (0.1075, 0.1075), (0.1446, 0.1446)]$$

$$DEF(N_{Z1}) = \frac{2 + 0.8 \cdot 0.8 - 0.15 \cdot 0.15 - 0.2 \cdot 0.2}{3} = 0.8592$$

Definition 3 [52]: Two weighted aggregation operators of neutrosophic Z-numbers.

Drawing upon Equation (7) and Equation (9) from Definition 2, we can formulate the weighted aggregation arithmetic mean (NZNWAA) equation for NZNs. Let $N_{Zi} = [\alpha_i(A, C), \beta_i(A, C), \gamma_i(A, C)] = [(\alpha_{Ai}, \alpha_{Ci}), (\beta_{Ai}, \beta_{Ci}), (\gamma_{Ai}, \gamma_{Ci})]$, ($i = 1, 2, \dots, n$) be a group of NZN and NZNWAA: $\Omega^n \rightarrow \Omega$. Subsequently, the NZNWAA equation is formally defined using Equation (12):

$$NZNWAA(N_{Z1}, N_{Z2}, \dots, N_{Zn}) = \sum_{i=1}^n \varepsilon_i N_{Zi} = \left[\left(1 - \prod_{i=1}^n (1 - \alpha_{Ai})^{\varepsilon_i}, 1 - \prod_{i=1}^n (1 - \alpha_{Ci})^{\varepsilon_i} \right), \left(\prod_{i=1}^n \beta_{Ai}^{\varepsilon_i}, \prod_{i=1}^n \beta_{Ci}^{\varepsilon_i} \right), \left(\prod_{i=1}^n \gamma_{Ai}^{\varepsilon_i}, \prod_{i=1}^n \gamma_{Ci}^{\varepsilon_i} \right) \right] \quad (12)$$

where $\varepsilon_i (i = 1, 2, \dots, n)$ is the weight of N_{Zi} with $0 \leq \varepsilon_i \leq 1$ and $\sum_{i=1}^n \varepsilon_i = 1$.

Likewise, leveraging Equation (8) and Equation (10) from Definition 2, we can derive the weighted aggregation geometric mean (NZNWAGM) equation for NZNs. Let $N_{Zi} = [\alpha_i(A, C), \beta_i(A, C), \gamma_i(A, C)] = [(\alpha_{Ai}, \alpha_{Ci}), (\beta_{Ai}, \beta_{Ci}), (\gamma_{Ai}, \gamma_{Ci})]$, ($i = 1, 2, \dots, n$) be a group of NZN and NZNWGA: $\Omega^n \rightarrow \Omega$. The NZNWGA equation is formally defined using Equation (13):

$$NZNWGA(N_{Z1}, N_{Z2}, \dots, N_{Zn}) = \prod_{i=1}^n (N_{Zi})^{\varepsilon_i} = \left[\left(\prod_{i=1}^n (\alpha_{Ai})^{\varepsilon_i}, \prod_{i=1}^n (\alpha_{Ci})^{\varepsilon_i} \right), \left(1 - \prod_{i=1}^n (1 - \beta_{Ai})^{\varepsilon_i}, 1 - \prod_{i=1}^n (1 - \beta_{Ci})^{\varepsilon_i} \right), \left(1 - \prod_{i=1}^n (1 - \gamma_{Ai})^{\varepsilon_i}, 1 - \prod_{i=1}^n (1 - \gamma_{Ci})^{\varepsilon_i} \right) \right] \quad (13)$$

where $\varepsilon_i (i = 1, 2, \dots, n)$ is the weight of N_{Zi} with $0 \leq \varepsilon_i \leq 1$ and $\sum_{i=1}^n \varepsilon_i = 1$.

Illustrative example 2: Assume a set of 4 NZN numbers: $\{[(0.8, 0.8), (0.15, 0.15), (0.2, 0.2)], [(0.6, 0.8), (0.35, 0.15), (0.4, 0.2)], [(0.4, 0.8), (0.65, 0.15), (0.6, 0.2)], [(0.2, 0.8), (0.85, 0.15), (0.8, 0.2)]\}$ with corresponding weights $\varepsilon_i = \{0.1, 0.3, 0.4, 0.2\}$. The agreed results using the NZNWAA (Equation (12)) and NZNWGA (Equation (13)) methods are as follows:

$$NZNWAA(N_{Z1}, N_{Z2}, N_{Z3}, N_{Z4}) = [(0.4958, 0.8), (0.4919, 0.15), (0.5042, 0.2)]$$

$$NZNWGA(N_{Z1}, N_{Z2}, N_{Z3}, N_{Z4}) = [(0.4215, 0.7999), (0.6112, 0.15), (0.5785, 0.2001)]$$

Definition 4 [59]: Distance and similarity measures of NZN sets.

In recent years, researchers have increasingly focused on distance and similarity metrics between sets, which serve as potent decision-making tools [76]. We can define the distance between two NZN sets and their respective weights. Let $N_{Z1} = \{N_{Z11}, N_{Z12}, \dots, N_{Z1n}\}$ and $N_{Z2} = \{N_{Z21}, N_{Z22}, \dots, N_{Z2n}\}$, where $N_{Z1k} = [\alpha_{1k}(A, C), \beta_{1k}(A, C), \gamma_{1k}(A, C)] = [(\alpha_{A1k}, \alpha_{C1k}), (\beta_{A1k}, \beta_{C1k}), (\gamma_{A1k}, \gamma_{C1k})]$, and $N_{Z2k} = [\alpha_{2k}(A, C), \beta_{2k}(A, C), \gamma_{2k}(A, C)] = [(\alpha_{A2k}, \alpha_{C2k}), (\beta_{A2k}, \beta_{C2k}), (\gamma_{A2k}, \gamma_{C2k})]$ are two NZNs, set $\vartheta \geq 1$ as any integer, and the corresponding weights of n pairs of NZN $w_k = (w_1, w_2, \dots, w_n)$, $\sum_{k=1}^n w_k = 1$. Subsequently, the generalized distance between N_{Z1} and N_{Z2} is calculated using Equation (14)

$$D_{w\vartheta}(N_{Z1}, N_{Z2}) = \frac{1}{2} \left\{ \sqrt[\vartheta]{\frac{1}{3} \sum_{k=1}^n w_k (|\alpha_{A1k} - \alpha_{A2k}|^\vartheta + |\beta_{A1k} - \beta_{A2k}|^\vartheta + |\gamma_{A1k} - \gamma_{A2k}|^\vartheta)} + \sqrt[\vartheta]{\frac{1}{3} \sum_{k=1}^n w_k (|\alpha_{C1k} - \alpha_{C2k}|^\vartheta + |\beta_{C1k} - \beta_{C2k}|^\vartheta + |\gamma_{C1k} - \gamma_{C2k}|^\vartheta)} \right\} \quad (14)$$

When $\vartheta = 1$, the above-generalized distance formula becomes the Hamming distance formula D_{w1} , using Equation (15):

$$D_{w1}(N_{Z1}, N_{Z2}) = \frac{1}{6} \left\{ \sum_{k=1}^n w_k (|\alpha_{A1k} - \alpha_{A2k}| + |\beta_{A1k} - \beta_{A2k}| + |\gamma_{A1k} - \gamma_{A2k}|) + \sum_{k=1}^n w_k (|\alpha_{C1k} - \alpha_{C2k}| + |\beta_{C1k} - \beta_{C2k}| + |\gamma_{C1k} - \gamma_{C2k}|) \right\} \quad (15)$$

When $\vartheta = 2$, the above-generalized distance formula becomes the Euclidean distance formula D_{w2} , using Equation (16):

$$D_{w2}(N_{Z1}, N_{Z2}) = \frac{1}{2} \left\{ \sqrt{\frac{1}{3} \sum_{k=1}^n w_k (|\alpha_{A1k} - \alpha_{A2k}|^2 + |\beta_{A1k} - \beta_{A2k}|^2 + |\gamma_{A1k} - \gamma_{A2k}|^2)} + \sqrt{\frac{1}{3} \sum_{k=1}^n w_k (|\alpha_{C1k} - \alpha_{C2k}|^2 + |\beta_{C1k} - \beta_{C2k}|^2 + |\gamma_{C1k} - \gamma_{C2k}|^2)} \right\} \quad (16)$$

Illustrative example 3: Assume a 2 set of 2 NZN numbers: $N_{z1} = \{[(0.8,0.8),(0.15,0.15),(0.2,0.2)], [(0.6,0.8),(0.35,0.15),(0.4,0.2)]\}$ $N_{z2} = \{[(0.6,0.4),(0.35,0.65),(0.4,0.6)], [(0.2,0.2),(0.85,0.85),(0.8,0.8)]\}$ with corresponding weights $e_i = (0.65,0.35)$. The Hamming distance and Euclidean distance between these two NZN sets are calculated using Equations (15) and (16):

Hamming distance

$$D_{w1}(N_{z1}, N_{z2}) = \frac{1}{6} \{ [0.65 \bullet (|0.8 - 0.6| + |0.15 - 0.35| + |0.2 - 0.4|) + 0.35 \bullet (|0.6 - 0.2| + |0.35 - 0.85| + |0.4 - 0.8|)] + [0.65 \bullet (|0.8 - 0.4| + |0.15 - 0.65| + |0.2 - 0.6|) + 0.35 \bullet (|0.8 - 0.2| + |0.15 - 0.85| + |0.2 - 0.8|)] \} = 0.3925$$

Euclidean distance

$$D_{w2}(N_{z1}, N_{z2}) = \frac{1}{2} \left\{ \sqrt{\frac{1}{3} [0.65 \bullet (|0.8 - 0.6|^2 + |0.15 - 0.35|^2 + |0.2 - 0.4|^2) + 0.35 \bullet (|0.6 - 0.2|^2 + |0.35 - 0.85|^2 + |0.4 - 0.8|^2)]} + \sqrt{\frac{1}{3} [0.65 \bullet (|0.8 - 0.4|^2 + |0.15 - 0.65|^2 + |0.2 - 0.6|^2) + 0.35 \bullet (|0.8 - 0.2|^2 + |0.15 - 0.85|^2 + |0.2 - 0.8|^2)]} \right\} = 0.4093$$

3.3. NZN DELPHI method

Assume k experts provide assessments for n factors. Each expert evaluates the significance of each factor using a linguistic scale, which is then converted to NZN numbers using NZN sets. Additionally, experts are weighted based on their education and years of experience. The calculation steps are outlined below.

Step 1 Calculate the weight of the expert.

Expert weights will be assessed using NZN numbers, comprising two components: A, representing the degree of evaluation based on the expert's experience and education, and C, indicating the degree of certainty based on the research team's knowledge about the expert. The two NZN numbers representing the expert rating based on years of experience and education will be aggregated using Equation (7) and converted into a crisp score using Equation (11). Table 4 outlines the expert-level assessment along with the corresponding linguistic scale [77].

For instance, Expert 1 holds a master's degree and possesses 20 years of professional experience. The authors are familiar with this expert, indicating a high level of certainty regarding the assessment (VH). Consequently, the weighted assessment based on experience for this expert will be (VH; VH), while based on qualifications, it will be (H; VH). These assessments are represented as NZN numbers as follows: $[(0.6,0.8),(0.35,0.15),(0.4,0.2)]$ and $[(0.8,0.8),(0.15,0.15),(0.2,0.2)]$

Two evaluations in the form of Fuzzy numbers will be combined using Equation (7). The result will be converted to the number of crips using Equation (11) into the evaluation value. In the example above, the evaluation value of expert one is as follows:

$$[(0.6, 0.8), (0.35, 0.15), (0.4, 0.2)] \oplus [(0.6, 0.8), (0.35, 0.15), (0.4, 0.2)] = [(0.92, 0.96), (0.0525, 0.0225), (0.08, 0.04)]$$

Using Equation (11) to converted NZN $[(0.92,0.96),(0.0525,0.0225),(0.08,0.04)]$ to crips score, the results obtained is 0,9596.

Calculate the evaluation value for k experts, obtaining k values EK: $ek_j = \{ek_1, ek_2, \dots, ek_k\}$. The weight of expert EW: $ew_j = \{ew_1, ew_2, \dots, ew_k\}$ is calculated as Equation (15) below:

$$ew_j = \frac{ek_j}{\sum_{j=1}^k ek_j} \quad (17)$$

Step 2 Build a weighted expert evaluation matrix

Experts will engage in evaluating the importance of n factors. Initial evaluation results are presented in linguistic form and converted to NZN numbers, forming a matrix $\otimes EM = [em_{ij}]_{n \times k}$, where n represents the number of factors and k denotes the number of experts. The linguistic scale for evaluation and corresponding NZN are depicted in Table 5 [77].

Table 4
Expert rating scale.

Education (A)	Experience (A)	Certainty (C)	Linguistic scale	Code	NZN
Doctor	Over 20 years	Very high	Very high	VH	(0.8,0.15,0.2)
Master	10–20 years	High	High	H	(0.6,0.35,0.4)
Bachelor	5–10 years	Medium	Medium	M	(0.4,0.65,0.6)
Under Bachelor	Under five years	Low	Low	L	(0.2,0.85,0.8)
		Very low	Very low	VL	(0,1,1)

The weighted expert evaluation matrix $\otimes EMW = [emw_{ij}]_{n \times k}$ is created using Equation (18) below.

$$emw_{ij} = em_{ij} \otimes ew_j \quad (18)$$

where $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, k$; $ew_j = \{ew_1, ew_2, \dots, ew_k\}$ is the expert's j of weight

Step 3 Calculate the threshold and validate factors

Each factor will undergo an assessment by k experts. Employ Equation (12) to synthesize these k assessments, yielding aggregated results for n factors in the form of NZN. Subsequently, apply Equation (11) to convert the ratings into crisp scores, thereby generating n EV values: $ev_i = \{ev_1, ev_2, \dots, ev_n\}$. The threshold value is calculated using Equation (19) below:

$$\delta = \frac{\sum_{i=1}^n ev_i}{n} \quad (19)$$

If value $ev_i \geq \delta$ then factor i is accepted. If value $ev_i < \delta$ then factor i is rejected

3.4. NZN DEMATEL model

Let us assume there are k experts with their corresponding expert weights ew , evaluating the mutual influence of n factors. Ratings are initially expressed in linguistic form and subsequently converted to NZN. The rating scale and corresponding NZN are presented in Table 6 [77].

After converting the assessments to NZN numbers, the data will be processed using the DEMATEL method. The calculation steps are presented below.

Step 1 Establishing the direct relationship matrix $\otimes D$.

The evaluations of the mutual influence of n factors (where factor i affects factor j) from k experts, denoted as d_{ij}^k are converted to NZN with their respective expert weights ew_t . These evaluations are then consolidated using Equation (12), resulting in the direct influence matrix $\otimes D = [\otimes d_{ij}]_{n \times n}$, while:

$$d_{ij} = \text{NZNWAA} \left(d_{ij}^1, d_{ij}^2, \dots, d_{ij}^k \right) = \sum_{t=1}^k ew_t d_{ij}^k \quad (20)$$

where $i = 1, 2, \dots, n$, $j = 1, 2, \dots, n$, $t = 1, 2, \dots, k$; $\otimes d_{ij} = \left[\left(d_{ij}^{\alpha A}, d_{ij}^{\alpha C} \right), \left(d_{ij}^{\beta A}, d_{ij}^{\beta C} \right), \left(d_{ij}^{\gamma A}, d_{ij}^{\gamma C} \right) \right]$. Here, the diagonal elements in the matrix are 0, i.e., $\otimes d_{ij} = 0$ (when $i = j$).

Step 2 Calculating the normalized direct relationship matrix $\otimes D^*$.

Matrix $\otimes D = [\otimes d_{ij}]_{n \times n}$ will be normalized into the matrix $\otimes D^* = [\otimes d_{ij}^*]_{n \times n}$ using Equation (21) below:

$$d_{ij}^* = (\theta_A, \theta_C).d_{ij} = \left[\left(\theta_A d_{ij}^{\alpha A}, \theta_C d_{ij}^{\alpha C} \right), \left(\theta_A d_{ij}^{\beta A}, \theta_C d_{ij}^{\beta C} \right), \left(\theta_A d_{ij}^{\gamma A}, \theta_C d_{ij}^{\gamma C} \right) \right] \quad (21)$$

where $\otimes d_{ij}^* = \left[\left(d_{ij}^{*\alpha A}, d_{ij}^{*\alpha C} \right), \left(d_{ij}^{*\beta A}, d_{ij}^{*\beta C} \right), \left(d_{ij}^{*\gamma A}, d_{ij}^{*\gamma C} \right) \right]$.

Table 5
NZN DELPHI Linguistic important scale.

Important level	Code	Membership		
		α	β	γ
Very low	VL	0	1	1
Low	L	0.2	0.85	0.8
Medium	M	0.4	0.65	0.6
High	H	0.6	0.35	0.4
Very high	VH	0.8	0.15	0.2

Table 6
NZN DEMATEL Linguistic influence scale.

Influence level	Code	Membership		
		α	β	γ
Equal influence	EI	0	1	1
Weak influence	WI	0.2	0.85	0.8
Fair influence	FI	0.4	0.65	0.6
Very influence	VI	0.6	0.35	0.4
Absolute influence	AI	0.8	0.15	0.2

$$\theta_A = \text{Min} \left\{ \frac{1}{\sum_{j=1}^n d_{ij}^{\alpha_A}}, \frac{1}{\sum_{j=1}^n d_{ij}^{\beta_A}}, \frac{1}{\sum_{j=1}^n d_{ij}^{\gamma_A}} \right\} \quad (22)$$

$$\theta_C = \text{Min} \left\{ \frac{1}{\sum_{j=1}^n d_{ij}^{\alpha_C}}, \frac{1}{\sum_{j=1}^n d_{ij}^{\beta_C}}, \frac{1}{\sum_{j=1}^n d_{ij}^{\gamma_C}} \right\}$$

Step 3 Generating the total influence matrix $\otimes T$

The normalized direct relationship matrix $\otimes T$ is computed into a comprehensive influence matrix. This integration encompasses both direct and indirect influence relationships, aggregating them across a spectrum from minimal impact to maximal influence, spanning from the power of one to the power of infinity. The procedure is presented below:

$$\otimes T = [\otimes t_{ij}]_{n \times n}, i, j = 1, 2, \dots, n \quad (23)$$

where $\otimes t_{ij} = \left[\left(t_{ij}^{\alpha_A}, t_{ij}^{\alpha_C} \right), \left(t_{ij}^{\beta_A}, t_{ij}^{\beta_C} \right), \left(t_{ij}^{\gamma_A}, t_{ij}^{\gamma_C} \right) \right]$.

$$\begin{aligned} \otimes T &= \otimes D^* + \otimes D^{*2} + \dots + \otimes D^{*\infty} \\ &= \otimes T(I + \otimes D^* + \otimes D^{*2} + \dots + \otimes D^{*\infty-1}) \\ &= \otimes D^*(I - \otimes D^{*\infty})(I - \otimes D^*)^{-1} = \otimes D^*(I - \otimes D^*)^{-1} \end{aligned} \quad (24)$$

where $\otimes T^{\infty} = [0]_{n \times n}$ and I is the identity matrix

Matrix $\otimes T$ elements in the form NZN use Equation (11) to convert to crips NZN to create a matrix $\otimes T^* = [\otimes t_{ij}^*]_{n \times n}$.

Step 4 Establishing an INRM to identify the mutual influence of development projects.

$\otimes r$ is calculated by adding up each column of the total influence matrix $\otimes T^*$. $\otimes c$ is calculated by adding up each row of the matrix $\otimes T^*$.

$$\otimes r = [\otimes r_i]_{n \times 1} = (\otimes r_1, \otimes r_2, \dots, \otimes r_i, \dots, \otimes r_n) \quad (25)$$

$$[\otimes r_i]_{n \times 1} = \left[\sum_{j=1}^n \otimes t_{ij}^* \right]_{n \times 1} \quad (26)$$

$$\otimes c = [\otimes c_i]_{1 \times n} = (\otimes c_1, \otimes c_2, \dots, \otimes c_i, \dots, \otimes c_n)^T \quad (27)$$

$$[\otimes c_i]_{1 \times n} = \left[\sum_{i=1}^n \otimes t_{ij}^* \right]_{1 \times n} = [\otimes c_i]_{n \times 1}^T \quad (28)$$

Note: "superscript T" is the transpose of the matrix.

The index of the strength of influences imparted and received is $\otimes r_i + \otimes c_i$. The net influence is represented by $\otimes r_i - \otimes c_i$. A higher $\otimes r_i + \otimes c_i$ Indicates that criterion i has a greater influence on the evaluation system. Indicator i significantly influence others if $\otimes r_i - \otimes c_i > 0$ (is positive). Indicator i is influenced by other indicators if $\otimes r_i - \otimes c_i < 0$ (is negative)

The indicator's overall effect on the assessment system is represented by $\otimes r_i + \otimes c_i$. Therefore, Equation (29) is used to construct an

indicator's impact weight.

$$\phi_i = \frac{(r_i + c_i)}{\sum_{i=1}^n (r_i + c_i)} \quad (29)$$

3.5. NZN COCOSO model

Consider k experts tasked with evaluating the impact of n strategies on approaching and resolving m factors. The rating scale and corresponding NZN are illustrated in Table 7 [77]:

After converting the assessments to NZN numbers, the data will be processed using the COCOSO method. The calculation steps are presented below.

Step 1 Calculated synthesize expert assessment matrix $\otimes F$.

The evaluations of the effectiveness of n strategies in resolving m factors by k experts denoted as e_{ij}^k along with their corresponding expert weights ew_t are synthesized into matrix $\otimes F = [\otimes f_{ij}]_{n \times m}$ using Equation (12) is shown below.

$$f_{ij} = \text{NZNAA} \left(f_{ij}^1, f_{ij}^2, \dots, f_{ij}^k \right) = \sum_{t=1}^k ew_t f_{ij}^k$$

where $i = 1, 2, \dots, n$, $j = 1, 2, \dots, m$, $t = 1, 2, \dots, k$; $\otimes f_{ij} = \left[\left(f_{ij}^{\alpha_A}, f_{ij}^{\alpha_C} \right), \left(f_{ij}^{\beta_A}, f_{ij}^{\beta_C} \right), \left(f_{ij}^{\gamma_A}, f_{ij}^{\gamma_C} \right) \right]$.

Step 2 Normalized matrix $\otimes F$ into matrix $\otimes F^*$.

Matrix $\otimes F = [\otimes f_{ij}]_{n \times m}$ will be normalized into the matrix $\otimes F^* = [\otimes f_{ij}^*]_{n \times m}$ using Equations (30) and (31) below:

$$f_{ij}^* = (\xi_A, \xi_C) \cdot s_{ij} = \left[\left(\xi_A f_{ij}^{\alpha_A}, \xi_C f_{ij}^{\alpha_C} \right), \left(\xi_A f_{ij}^{\beta_A}, \xi_C f_{ij}^{\beta_C} \right), \left(\xi_A f_{ij}^{\gamma_A}, \xi_C f_{ij}^{\gamma_C} \right) \right] \quad (30)$$

where $\otimes f_{ij}^* = \left[\left(f_{ij}^{*\alpha_A}, s_{ij}^{*\alpha_C} \right), \left(f_{ij}^{*\beta_A}, f_{ij}^{*\beta_C} \right), \left(f_{ij}^{*\gamma_A}, f_{ij}^{*\gamma_C} \right) \right]$.

$$\xi_A = \text{Min} \left\{ \frac{1}{\sqrt{\sum_{j=1}^m \left(f_{ij}^{\alpha_A} \right)^2}}, \frac{1}{\sqrt{\sum_{j=1}^m \left(f_{ij}^{\beta_A} \right)^2}}, \frac{1}{\sqrt{\sum_{j=1}^m \left(f_{ij}^{\gamma_A} \right)^2}} \right\} \quad (31)$$

$$\xi_C = \text{Min} \left\{ \frac{1}{\sqrt{\sum_{j=1}^m \left(f_{ij}^{\alpha_C} \right)^2}}, \frac{1}{\sqrt{\sum_{j=1}^m \left(f_{ij}^{\beta_C} \right)^2}}, \frac{1}{\sqrt{\sum_{j=1}^m \left(f_{ij}^{\gamma_C} \right)^2}} \right\}$$

Step 3 Calculate the relative importance of each strategy

The relative importance of the given strategies is computed through the Weighted Sum Method Λ_i and the Weighted Product Method Ψ_i using Equations (9) and (10), respectively. The detailed formula is shown in Equations (31) and (32):

Table 7
NZN COCOSO Linguistic important scale.

Important level	Code	Membership		
		α	β	γ
Very Poor	VP	0	1	1
Poor	P	0.2	0.85	0.8
Fair	F	0.4	0.65	0.6
Good	G	0.6	0.35	0.4
Very Good	VG	0.8	0.15	0.2

$$\Lambda_i = \sum_j^m \phi_j f_{ij}^* \quad (31a)$$

$$\Psi_i = \sum_j^m \left(f_{ij}^*\right)^{\phi_j} \quad (32)$$

where $= 1, 2, \dots, n$, $j = 1, 2, \dots, m$, $\phi_j = (\phi_1, \phi_2, \dots, \phi_m)$ is the weight of factor j

Step 4 Calculate the relative weight of each strategy

In this step, three appraisal score $\mathbb{C}_{II}$, $\mathbb{C}_{III}$, $\mathbb{C}_{III}$ strategies are employed to determine the relative weights of other options, which are derived using the following formulas:

Calculate $\mathbb{C}_{II}$.

Using Equation (7) to sum up Λ_i and Ψ_i , after that, use Equation (11) to convert the result into a crisp score. Λ_i and Ψ_i Also are converted to crisp score by using Equation (11). The formula for calculating $\mathbb{C}_{II}$ is presented in Equation (33):

$$\mathbb{C}_{II} = \frac{DEF(\Lambda_i \oplus \Psi_i)}{\sum_{i=1}^n (DEF(\Lambda_i) + DEF(\Psi_i))} \quad (33)$$

Calculate $\mathbb{C}_{III}$.

$\mathbb{C}_{III}$ is calculated using Equation (37). Equation (37) uses the formula for dividing two numbers NZN. The formula for dividing 2 NZN numbers mentioned by Yazdani et al. [75] is as Equation (34) follows:

Let $N_{Z1} = [\alpha_1(A, C), \beta_1(A, C), \gamma_1(A, C)] = [(\alpha_{A1}, \alpha_{C1}), (\beta_{A1}, \beta_{C1}), (\gamma_{A1}, \gamma_{C1})]$ and $N_{Z2} = [\alpha_2(A, C), \beta_2(A, C), \gamma_2(A, C)] = [(\alpha_{A2}, \alpha_{C2}), (\beta_{A2}, \beta_{C2}), (\gamma_{A2}, \gamma_{C2})]$ be two NZNs, we have:

$$N_{Z1} \oslash N_{Z2} = \left[\left(\frac{\alpha_{A1}}{\alpha_{A2}}, \frac{\alpha_{C1}}{\alpha_{C2}} \right), \left(\frac{\beta_{A1}}{\beta_{A2}}, \frac{\beta_{C1}}{\beta_{C2}} \right), \left(\frac{\gamma_{A1}}{\gamma_{A2}}, \frac{\gamma_{C1}}{\gamma_{C2}} \right) \right] \quad (34)$$

$\Lambda_{\min i}$ and $\Psi_{\min i}$ is determine by using Equation (35) and (36) below [59]:

$$\Lambda_{\min i} = \left[\left(\min\{\Lambda_i^{\alpha_A}\}, \min\{\Lambda_i^{\alpha_C}\} \right), \left(\max\{\Lambda_i^{\beta_A}\}, \max\{\Lambda_i^{\beta_C}\} \right), \left(\max\{\Lambda_i^{\gamma_A}\}, \max\{\Lambda_i^{\gamma_C}\} \right) \right] = \left[\left(\Lambda_i^{\alpha_A}, \Lambda_i^{\alpha_C} \right), \left(\Lambda_i^{\beta_A}, \Lambda_i^{\beta_C} \right), \left(\Lambda_i^{\gamma_A}, \Lambda_i^{\gamma_C} \right) \right] \quad (35)$$

$$\Psi_{\min i} = \left[\left(\min\{\Psi_i^{\alpha_A}\}, \min\{\Psi_i^{\alpha_C}\} \right), \left(\max\{\Psi_i^{\beta_A}\}, \max\{\Psi_i^{\beta_C}\} \right), \left(\max\{\Psi_i^{\gamma_A}\}, \max\{\Psi_i^{\gamma_C}\} \right) \right] = \left[\left(\Psi_i^{\alpha_A}, \Psi_i^{\alpha_C} \right), \left(\Psi_i^{\beta_A}, \Psi_i^{\beta_C} \right), \left(\Psi_i^{\gamma_A}, \Psi_i^{\gamma_C} \right) \right] \quad (36)$$

Apply Equation (34)–(36) to Equation (37), $\mathbb{C}_{III}$ is computed as follows:

$$\mathbb{C}_{III} = DEF\{0.5 \bullet [(\Lambda_i \oslash \Lambda_{\min i}) \oplus (\Psi_i \oslash \Psi_{\min i})]\} \quad (37)$$

Calculate $\mathbb{C}_{III}$.

Equation (38) and (39) is used to determine $\Lambda_{\max i}$ and $\Psi_{\max i}$ as follows [59]:

$$\Lambda_{\max i} = \left[\left(\max\{\Lambda_i^{\alpha_A}\}, \max\{\Lambda_i^{\alpha_C}\} \right), \left(\min\{\Lambda_i^{\beta_A}\}, \min\{\Lambda_i^{\beta_C}\} \right), \left(\min\{\Lambda_i^{\gamma_A}\}, \min\{\Lambda_i^{\gamma_C}\} \right) \right] = \left[\left(\Lambda_i^{\alpha_A}, \Lambda_i^{\alpha_C} \right), \left(\Lambda_i^{\beta_A}, \Lambda_i^{\beta_C} \right), \left(\Lambda_i^{\gamma_A}, \Lambda_i^{\gamma_C} \right) \right] \quad (38)$$

$$\Psi_{\max i} = \left[\left(\max\{\Psi_i^{\alpha_A}\}, \max\{\Psi_i^{\alpha_C}\} \right), \left(\min\{\Psi_i^{\beta_A}\}, \min\{\Psi_i^{\beta_C}\} \right), \left(\min\{\Psi_i^{\gamma_A}\}, \min\{\Psi_i^{\gamma_C}\} \right) \right] = \left[\left(\Psi_i^{\alpha_A}, \Psi_i^{\alpha_C} \right), \left(\Psi_i^{\beta_A}, \Psi_i^{\beta_C} \right), \left(\Psi_i^{\gamma_A}, \Psi_i^{\gamma_C} \right) \right] \quad (39)$$

With $0 \leq \varpi \leq 1$, Λ_i , Ψ_i , $\Lambda_{\max i}$ and $\Psi_{\max i}$ is converted into a crisp score by using Equation (11). Apply to Equation (40) to calculate $\mathbb{C}_{III}$ as below:

$$\mathbb{C}_{III} = \frac{\varpi DEF(\Lambda_i) + (1 - \varpi)DEF(\Psi_i)}{\varpi DEF(\Lambda_{\max i}) + (1 - \varpi)DEF(\Psi_{\max i})} \quad (40)$$

ϖ is usually taken as 0.5 [75]. However, this study will set ϖ values from 0.1, 0.2, to 0.9 in turn to do sensitive analysis.

Calculate $\mathbb{C}_i$ and ranking

$\mathbb{C}_i$ is calculated using Equation (41) below and the $\mathbb{C}_i$ value is used to rank the corresponding strategies.

$$\mathbb{C}_i = \sqrt[3]{\mathbb{C}_{II}\mathbb{C}_{III}\mathbb{C}_{III}} + \frac{1}{3}(\mathbb{C}_{II} + \mathbb{C}_{III} + \mathbb{C}_{III}) \quad (41)$$

4. Case study

4.1. Problem Description

Cybersecurity risks within the finance and banking sector have become a focal point of research and concern, drawing attention not only from financial institutions but also from governmental bodies overseeing management and operations, as well as from consumers themselves [78]. According to statistics provided by the Ministry of Information and Communications of Vietnam, during the first 11 months of 2023, nearly 16,000 reports of fraudulent activities were submitted by Vietnamese Internet users through alert systems. Alarming, over 91 % of these reports were linked to fraud and illicit activities within the banking and financial domain [79]. The prevalence of cybersecurity threats poses significant challenges to the integrity, security, and operational efficiency of the banking and financial infrastructure. Such risks extend beyond financial institutions, impacting individuals, businesses, and governmental entities that rely on these systems for their operations [8]. The ramifications of cybersecurity breaches can result in substantial economic losses, making it imperative for all stakeholders to prioritize robust security measures and proactive risk management strategies.

This study aims to analyze cybersecurity risks within Vietnam's finance and banking system, assessing their impact and interrelations. It also proposes and prioritizes solutions based on expert evaluations of their efficacy in risk mitigation. By identifying, quantifying, and understanding these risks, the study seeks to enhance the resilience and security of the financial infrastructure while minimizing potential economic losses and disruptions. Fig. 2 illustrates the hierarchy of cybersecurity risks impacting Vietnam's financial and banking system for a comprehensive overview.

4.2. Expert selection and Expert weight

Thirty experts, selected for their expertise and experience pertinent to the scope of this study, participated in rating the importance of those above seventeen potential risks. These experts encompass individuals who serve as educators, researchers, and practitioners directly involved in the fields of banking and finance, information security, and financial information systems. All experts possess over ten years of experience in the field, with nine experts boasting more than 20 years of expertise. Moreover, 93 % of the experts hold a

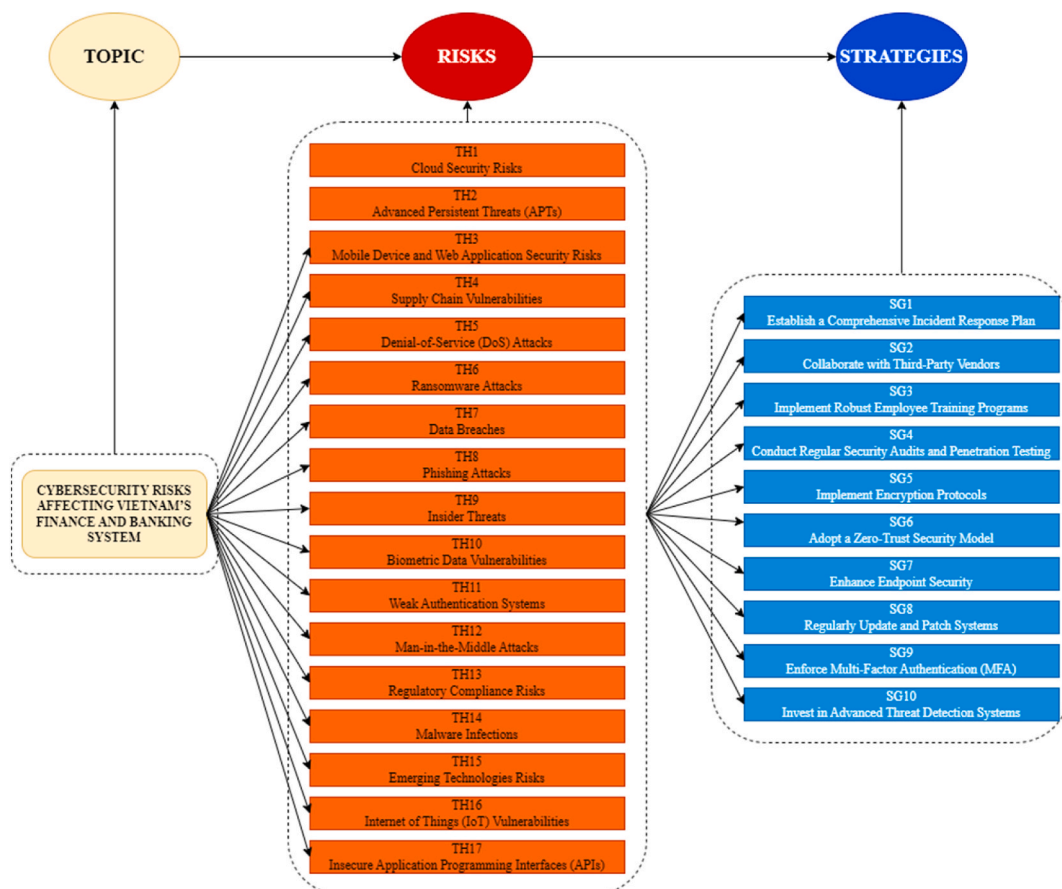


Fig. 2. Hierarchical framework of cybersecurity risks and strategies.

Master's degree or higher, with 50 % obtaining a doctoral degree. As outlined in section 3.2, the weight of each expert is estimated based on their years of experience and educational attainment, as shown in Table 4. The profiles of each expert, along with their corresponding weights, are detailed in Table 8.

4.3. NZN DELPHI results

The experts assessed the relevance and importance of each risk to Vietnam's finance and banking system using the evaluation scale presented in Table 5. The evaluation results indicate that TH13 - Regulatory Compliance Risks and TH15 - Emerging Technologies Risks were rejected. A detailed presentation of the evaluation results for the 17 risks is provided in Table 9 below.

4.4. NZN DEMATEL results

After the NZN DELPHI phase, fifteen validated risks were included for evaluating their impact weight on the finance and banking system in Vietnam, as well as the cause-and-effect relationships between these factors. As outlined in section 3.3, experts evaluated pairs of factors to assess how they influence each other using the evaluation scale presented in Table 6. The data was processed using the DEMATEL method, presenting the total influence matrix as NZN numbers, as depicted in Table 10 below.

The results presented in Table 10 were processed using Equation (12). Subsequently, Equations (25)–(29) were applied to calculate the values of $\otimes r$, $\otimes c$, $\otimes r + \otimes c$, $\otimes r - \otimes c$, and the corresponding risk weights. The research findings indicate that eight risks were classified into the cause group, exhibiting positive $\otimes r - \otimes c$ values: TH3, TH4, TH9, TH10, TH11, TH14, TH16, and TH17. Conversely, the remaining seven risks were categorized into the effect group, namely TH1, TH2, TH5, TH6, TH7, TH8, and TH12. The weights and impact ratings of these risks are detailed in Table 11:

Table 11 reveals that TH16 - Internet of Things (IoT) Vulnerabilities and TH17 - Insecure Application Programming Interfaces (APIs) exhibit the greatest net influence, indicating their substantial causal impact. TH16 - Internet of Things (IoT) Vulnerabilities, particularly highlighted by Javaheri et al. [13], emphasize significant risks related to IoT, potentially leading to Dos attacks or malware intrusion. Vulnerable IoT devices within banking infrastructure can be exploited, compromising sensitive financial data or launching cyberattacks. While enhancing convenience, the proliferation of IoT devices also escalates cybersecurity threats as cybercriminals exploit device connectivity for attacks [80]. However, safeguarding IoT devices requires frequent and resource-intensive updates. TH17 - Insecure Application Programming Interfaces (APIs) ranks second in terms of their net impact value. Vulnerabilities within APIs connecting different banking systems pose significant risks, allowing cybercriminals to exploit security weaknesses to gain unauthorized access to sensitive data or manipulate transactions [81]. Weaknesses in API security protocols may enable interception of

Table 8
Expert's profiles and expert weight.

Expert	Education	Experience	Evaluation value (NZN)	Evaluation value (Crips)	Expert Weight
Expert 1	Master	Over 20 years	[(0.92,0.96),(0.0525,0.0225),(0.08,0.04)]	0.9596	0.0408
Expert 2	Master	From 10 to 20 years	[(0.84,0.64),(0.1225,0.4225),(0.16,0.36)]	0.8094	0.0344
Expert 3	Doctor	From 10 to 20 years	[(0.92,0.36),(0.0525,0.7225),(0.08,0.64)]	0.7474	0.0317
Expert 4	Master	From 10 to 20 years	[(0.84,0),(0.1225,1),(0.16,1)]	0.5725	0.0243
Expert 5	Master	From 10 to 20 years	[(0.84,0.36),(0.1225,0.7225),(0.16,0.64)]	0.7038	0.0299
Expert 6	Master	From 10 to 20 years	[(0.84,0),(0.1225,1),(0.16,1)]	0.5725	0.0243
Expert 7	Master	From 10 to 20 years	[(0.84,0.64),(0.1225,0.4225),(0.16,0.36)]	0.8094	0.0344
Expert 8	Master	From 10 to 20 years	[(0.84,0.96),(0.1225,0.0225),(0.16,0.04)]	0.9324	0.0396
Expert 9	Master	From 10 to 20 years	[(0.84,0.64),(0.1225,0.4225),(0.16,0.36)]	0.8094	0.0344
Expert 10	Master	Over 20 years	[(0.92,0.64),(0.0525,0.4225),(0.08,0.36)]	0.8459	0.0359
Expert 11	Master	Over 20 years	[(0.92,0.36),(0.0525,0.7225),(0.08,0.64)]	0.7474	0.0317
Expert 12	Master	Over 20 years	[(0.92,0.84),(0.0525,0.1225),(0.08,0.16)]	0.9179	0.039
Expert 13	Master	Over 20 years	[(0.92,0),(0.0525,1),(0.08,1)]	0.6225	0.0264
Expert 14	Master	Over 20 years	[(0.92,0.96),(0.0525,0.0225),(0.08,0.04)]	0.9596	0.0408
Expert 15	Doctor	From 10 to 20 years	[(0.92,0),(0.0525,1),(0.08,1)]	0.6225	0.0264
Expert 16	Master	From 10 to 20 years	[(0.84,0.84),(0.1225,0.1225),(0.16,0.16)]	0.8883	0.0377
Expert 17	Doctor	From 10 to 20 years	[(0.92,0.96),(0.1225,0.0225),(0.08,0.04)]	0.9596	0.0408
Expert 18	Doctor	From 10 to 20 years	[(0.92,0),(0.0525,1),(0.08,1)]	0.6225	0.0264
Expert 19	Master	From 10 to 20 years	[(0.84,0),(0.1225,1),(0.16,1)]	0.5725	0.0243
Expert 20	Master	From 10 to 20 years	[(0.84,0.64),(0.1225,0.4225),(0.16,0.36)]	0.8094	0.0344
Expert 21	Master	From 10 to 20 years	[(0.84,0),(0.1225,1),(0.16,1)]	0.5725	0.0243
Expert 22	Master	From 10 to 20 years	[(0.84,0.36),(0.1225,0.7225),(0.16,0.64)]	0.7038	0.0299
Expert 23	Doctor	From 10 to 20 years	[(0.92,0.64),(0.0525,0.4225),(0.08,0.36)]	0.8459	0.0359
Expert 24	Master	From 10 to 20 years	[(0.84,0.96),(0.1225,0.0225),(0.16,0.04)]	0.9324	0.0396
Expert 25	Doctor	From 10 to 20 years	[(0.92,0.36),(0.0525,0.7225),(0.08,0.64)]	0.7474	0.0317
Expert 26	Bachelor	From 10 to 20 years	[(0.76,0.96),(0.2275,0.0225),(0.24,0.04)]	0.905	0.0384
Expert 27	Master	From 10 to 20 years	[(0.84,0.96),(0.1225,0.0225),(0.16,0.04)]	0.9324	0.0396
Expert 28	Master	From 10 to 20 years	[(0.84,0),(0.1225,1),(0.16,1)]	0.5725	0.0243
Expert 29	Doctor	From 10 to 20 years	[(0.92,0.96),(0.0525,0.0225),(0.08,0.04)]	0.9596	0.0408
Expert 30	Master	From 10 to 20 years	[(0.84,0.84),(0.1225,0.1225),(0.16,0.16)]	0.8883	0.0377
Sum evaluation value				23,5443	

Table 9
NZN DELPHI results.

Risks	Aggregate 30 assessment	Crips value	Validate
TH1	[(0.6506,0.5646),(0.3165,0.4176),(0.3494,0.4354)]	0.6943	Accepted
TH2	[(0.604,0.5491),(0.3527,0.4356),(0.396,0.4509)]	0.6665	Accepted
TH3	[(0.535,0.6367),(0.4319,0.3347),(0.465,0.3633)]	0.6757	Accepted
TH4	[(0.6437,0.6093),(0.3119,0.3678),(0.3563,0.3907)]	0.7128	Accepted
TH5	[(0.6486,0.5846),(0.3112,0.3923),(0.3514,0.4154)]	0.7037	Accepted
TH6	[(0.5876,0.5765),(0.3774,0.4036),(0.4124,0.4235)]	0.6706	Accepted
TH7	[(0.4928,0.6545),(0.4751,0.3106),(0.5072,0.3455)]	0.6666	Accepted
TH8	[(0.5945,0.647),(0.3673,0.3233),(0.4055,0.353)]	0.7076	Accepted
TH9	[(0.6041,0.6527),(0.3526,0.3136),(0.3959,0.3473)]	0.7154	Accepted
TH10	[(0.5176,0.6601),(0.4631,0.3031),(0.4824,0.3399)]	0.6791	Accepted
TH11	[(0.5458,0.664),(0.4233,0.2993),(0.4542,0.336)]	0.6944	Accepted
TH12	[(0.5707,0.6122),(0.3944,0.3618),(0.4293,0.3878)]	0.6801	Accepted
TH13	[(0.4824,0.3313),(0.5106,0.6911),(0.5176,0.6687)]	0.4869	Rejected
TH14	[(0.5137,0.635),(0.4592,0.3385),(0.4863,0.365)]	0.6644	Accepted
TH15	[(0.4889,0.3403),(0.4946,0.696),(0.5111,0.6597)]	0.495	Rejected
TH16	[(0.6026,0.5809),(0.3589,0.3921),(0.3974,0.4191)]	0.6809	Accepted
TH17	[(0.5296,0.6446),(0.4373,0.3218),(0.4704,0.3554)]	0.6778	Accepted
Threshold		0.663	

critical data transmitted between banking systems, including customer account information and transaction details. Cybercriminals can manipulate API endpoints to execute unauthorized transactions or tamper with data within banking systems. As highlighted by previous research by Al-Rumaim and Pawar [63], these vulnerabilities contribute to an increased risk of cyber-attacks and compromise the system's integrity. Considering the top 5 risks with the highest influence weight, as presented in Table 11, these findings indicate that TH14 - Malware Infections is the cybersecurity risk exerting the most significant impact on Vietnam's banking and financial system. This outcome aligns with the study of Shulha et al. [8] and Javaheri et al. [13], highlighting the substantial impact of malware infections on the efficiency of information technology systems and banking operations. Indeed, malware poses a grave threat to the integrity of banking systems, potentially resulting in unauthorized transactions, data breaches, and disruptions to banking services. Javaheri et al. [13] have mentioned that the financial services sector accounted for 19 % of total malware attacks and incurred approximately \$18.3 billion in losses in 2017 alone, making it the primary target of malware perpetrators. Ranked in second place is TH4 - Supply Chain Vulnerabilities. Previous research by Ahmed et al. [10] also highlighted Supply Chain Vulnerabilities as one of the top risks affecting the banking system. Undoubtedly, vulnerabilities in the security protocols of third-party vendors can be exploited, resulting in unauthorized access to financial systems and sensitive data [82]. The financial system's integrity is jeopardized when third-party hardware, servers, or transmission line providers suffer from issues such as aging infrastructure, subpar quality, lack of synchronization, or inadequate security measures. Consequently, any cyber-attacks targeting third-party devices, services, and systems can swiftly impact the financial and banking system. TH9 - Insider Threats is assessed to have the third-highest level of influence on Vietnam's financial and banking system. According to Javaheri et al. [13], human-related risks or insider attacks are prevalent cybersecurity concerns affecting the financial system. With a 47 % increase in just two years, insider cybersecurity threats are evidently becoming a progressively serious issue for financial and credit institutions [83]. This risk entails employees or individuals with privileged access posing a threat by intentionally or unintentionally causing harm. Insider threats entail the risk of data theft, fraud, or sabotage by individuals with access to sensitive financial information. This results in significant economic damage and undermines customer trust in the financial system, impacting its integrity. Ranked fourth is TH11 - Weak Authentication Systems, which highlights security and authentication policies within the system. Inadequate authentication methods can make accessing customers' accounts and sensitive financial information easier for cybercriminals. A weak authentication system heightens the risk of unauthorized access, particularly in the age of advanced artificial intelligence technologies capable of mimicking biological characteristics. Javaheri et al. [13] also emphasize authentication as a critical area for addressing cybersecurity risks. Ranked fifth among the top five risks with the highest influence is TH3 - Mobile Device and Web Application Security Risks. Shulha et al. [8] have mentioned the significant impact of attacks on vulnerabilities in web application systems on banking information security. Mobile banking and web application vulnerabilities can be exploited to steal users' credentials and conduct fraudulent transactions. Data from security firm Kaspersky in July 2023 reveals that 17,847,857 malicious emails were blocked by Kaspersky's Anti-Phishing system in 2022, with 1,569,005 attacks targeting businesses and 16,278,852 targeting consumers in Vietnam, especially on mobile devices [84]. Consequently, vulnerabilities in mobile and web applications pose a growing risk that profoundly impacts the security of the financial and banking system and users in Vietnam.

Considering the cause-and-effect relationship between risks, Fig. 3 below is an influential network relation map (INRM) that illustrates the relationship between factors. In the INRM, the arrow symbol denotes the direction of interaction between risks. Only relationships with an influence value exceeding the average value are depicted by arrows in the total influence matrix. By taking the value $\otimes r - \otimes c = 0$ as the horizontal axis and the average value of the most significant weight value and the smallest weight value as the secondary vertical axis, we can delineate four risk groups across four quadrants, as shown: Quadrant I comprises risks belonging to the cause group with strong influence, namely TH14, TH4, TH9, TH11, and TH3. Quadrant II includes cause group factors with a lower influence compared to those in Quadrant I, consisting of TH16, TH10, and TH17. However, the net influence value of these two risks TH16 and TH17 is the largest, so it is very noteworthy. Quadrant IV encompasses risks belonging to the effect group with significant

Table 10

Total influence matrix.

Risks	TH1	TH2	TH3	TH4	TH5	TH6	TH7	TH8	TH9	TH10	TH11	TH12	TH14	TH16	TH17	
TH1	[[0.4177, 1.1331], 0.1372,0 079],(0.1 969,0.10 92)]	[[0.4643, 1.1986], 0.1936,0 1313],(0.1 2572,0.1 64)]	[[0.4404, 1.162],(0 2033,0.1 371),(0.2 701),(0.1 31),(0.17)]	[[0.449,1 2074],(0 1.216), 0.1855,0.1 0.1854,0 301),(0.2 1404), 245,0.17 19)]	[[0.4801, 1.216], 0.1855,0.1 0.1854,0 301),(0.2 1404), 245,0.17 19)]	[[0.4853, 1.1737], 0.1872,0 0.1872,0 1297],(0.1 2492,0.1 2383,0.1 2683,0.1 2691,0.1 578)]	[[0.4806, 1.2005], 0.1872,0 0.1872,0 1297],(0.1 2492,0.1 2383,0.1 2683,0.1 2691,0.1 578)]	[[0.4923, 1.2455], 0.1767,0 0.1767,0 1297],(0.1 2492,0.1 2383,0.1 2683,0.1 2691,0.1 578)]	[[0.4468, 1.2299], 0.1895,0 0.1895,0 1297],(0.1 2492,0.1 2383,0.1 2683,0.1 2691,0.1 578)]	[[0.4438, 1.1895], 0.1895,0 0.1895,0 1297],(0.1 2492,0.1 2383,0.1 2683,0.1 2691,0.1 578)]	[[0.4744, 1.1974], 0.1895,0 0.1895,0 1297],(0.1 2492,0.1 2383,0.1 2683,0.1 2691,0.1 578)]	[[0.4538, 1.2187], 0.1912,0 0.1912,0 1297],(0.1 2492,0.1 2383,0.1 2683,0.1 2691,0.1 578)]	[[0.4701, 1.2146], 0.1912,0 0.1912,0 1297],(0.1 2492,0.1 2383,0.1 2683,0.1 2691,0.1 578)]	[[0.3658, 1.1787], 0.2553,0 0.2553,0 1297],(0.1 2492,0.1 2383,0.1 2683,0.1 2691,0.1 578)]	[[0.3566, 1.2191], 0.2532,0 0.2532,0 1297],(0.1 2492,0.1 2383,0.1 2683,0.1 2691,0.1 578)]	
TH2	[[0.4387, 1.2328], 0.2144,0 1268],(0.1 2744,0.1 596)]	[[0.382,1 1916],(0 0.23,0.12 42), 0.735],(0.2 12), 0.41562], 4)]	[[0.4059, 1.2153], 0.23,0.12 42), 0.41562], 4)]	[[0.4205, 1.2613], 0.23,0.12 42), 0.41562], 4)]	[[0.4631, 1.2741], 0.23,0.12 42), 0.41562], 4)]	[[0.4563, 1.2315], 0.23,0.12 42), 0.41562], 4)]	[[0.4591, 1.2529], 0.23,0.12 42), 0.41562], 4)]	[[0.4625, 1.2809], 0.23,0.12 42), 0.41562], 4)]	[[0.4625, 1.2809], 0.23,0.12 42), 0.41562], 4)]	[[0.4248, 1.2742], 0.23,0.12 42), 0.41562], 4)]	[[0.4139, 1.2344], 0.23,0.12 42), 0.41562], 4)]	[[0.4384, 1.2514], 0.23,0.12 42), 0.41562], 4)]	[[0.4324, 1.2609], 0.23,0.12 42), 0.41562], 4)]	[[0.4356, 1.2619], 0.23,0.12 42), 0.41562], 4)]	[[0.3539, 1.216], 0.23,0.12 42), 0.41562], 4)]	[[0.3464, 1.2693], 0.23,0.12 42), 0.41562], 4)]
TH3	[[0.5795, 1.3355], 0.1355,0 1063],(0.1 1936,0.1 393)]	[[0.5489, 1.3468], 0.1587,0 1082],(0.1 2149,0.1 406)]	[[0.4574, 1.2456], 0.118,0.0 684],(0.1 745,0.09 68)]	[[0.5465, 1.364],(0 1.502,0.0 1.0334,0 1028],(0.1 1901,0.1 1847,0.1 372)]	[[0.5852, 1.3694], 0.1334,0 1.0285,0 1047],(0.1 1847,0.1 1847,0.1 372)]	[[0.5946, 1.3301], 0.1334,0 1.0285,0 1047],(0.1 1847,0.1 1847,0.1 372)]	[[0.5848, 1.3485], 0.1334,0 1.0285,0 1047],(0.1 1847,0.1 1847,0.1 372)]	[[0.5905, 1.3875], 0.1334,0 1.0285,0 1047],(0.1 1847,0.1 1847,0.1 372)]	[[0.5456, 1.3855], 0.1334,0 1.0285,0 1047],(0.1 1847,0.1 1847,0.1 372)]	[[0.5335, 1.3348], 0.1334,0 1.0285,0 1047],(0.1 1847,0.1 1847,0.1 372)]	[[0.561, 1.3484], 0.1334,0 1.0285,0 1047],(0.1 1847,0.1 1847,0.1 372)]	[[0.5351, 1.3622], 0.1334,0 1.0285,0 1047],(0.1 1847,0.1 1847,0.1 372)]	[[0.5645, 1.3534], 0.1334,0 1.0285,0 1047],(0.1 1847,0.1 1847,0.1 372)]	[[0.451,1 1.3118], 0.1334,0 1.0285,0 1047],(0.1 1847,0.1 1847,0.1 372)]	[[0.433,1 1.3578], 0.1334,0 1.0285,0 1047],(0.1 1847,0.1 1847,0.1 372)]	
TH4	[[0.5521, 1.3645], 0.1507,0 1027],(0.1 2103,0.1 356)]	[[0.5393, 1.3844], 0.1551,0 1082],(0.1 2149,0.1 406)]	[[0.4977, 1.3366], 0.1761,0 1077],(0.1 2383,0.1 399)]	[[0.4599, 1.3249], 0.118,0.0 607],(0.1 732,0.08 83)]	[[0.5628, 1.3991], 0.1446,0 1.002],(0.1 2021,0.1 315)]	[[0.5677, 1.3584], 0.1446,0 1.002],(0.1 2021,0.1 315)]	[[0.5723, 1.375], 0.1406,0 0.953],(0.1 954,0.13 51)]	[[0.5723, 1.4225], 0.1406,0 0.953],(0.1 954,0.13 51)]	[[0.5723, 1.4225], 0.1406,0 0.953],(0.1 954,0.13 51)]	[[0.5291, 1.415], 0.1406,0 0.953],(0.1 954,0.13 51)]	[[0.5127, 1.363], 0.1514,0 1.02],(0.2 2225,0.1 121,0.13 317)]	[[0.5265, 1.3928], 0.1616,0 1.02],(0.2 2225,0.1 121,0.13 317)]	[[0.5381, 1.3885], 0.1644,0 1.02],(0.2 2225,0.1 121,0.13 317)]	[[0.4417, 1.3397], 0.1945,0 1.02],(0.2 2225,0.1 121,0.13 317)]	[[0.4155, 1.3953], 0.2076,0 1.02],(0.2 2225,0.1 121,0.13 317)]	
TH5	[[0.4827, 1.3101], 0.1808,0 1128],(0.1 2449,0.1 448)]	[[0.4617, 1.3234], 0.1981,0 1117],(0.1 2611,0.1 435)]	[[0.4384, 1.2852], 0.2064,0 1155],(0.1 2731,0.1 472)]	[[0.4529, 1.3362], 0.2004,0 1081],(0.1 2636,0.1 4)]	[[0.4259, 1.2811], 0.1311,0 0.654],(0.1 1897,0.0 932)]	[[0.4259, 1.301], 0.1311,0 0.654],(0.1 1897,0.0 932)]	[[0.4828, 1.3455], 0.1709,0 1.054],(0.1 2442,0.1 34)]	[[0.4828, 1.3455], 0.1709,0 1.054],(0.1 2442,0.1 34)]	[[0.4828, 1.3455], 0.1709,0 1.054],(0.1 2442,0.1 34)]	[[0.4451, 1.3588], 0.2072,0 1.044],(0.1 2709,0.1 364)]	[[0.4315, 1.3223], 0.2207,0 1.091],(0.1 2828,0.1 454)]	[[0.4633, 1.3223], 0.2207,0 1.091],(0.1 2828,0.1 454)]	[[0.4517, 1.3389], 0.2012,0 1.064],(0.1 259,0.14 391)]	[[0.4826, 1.3424], 0.178,0.0 989],(0.2 429,0.13 22)]	[[0.3802, 1.2908], 0.24,0.11 59],(0.30 91,0.147 9)]	
TH6	[[0.4772, 1.3159], 0.182,0.1 091],(0.2 469,0.14 17)]	[[0.4591, 1.3267], 0.1979,0 1112],(0.1 2603,0.1 431)]	[[0.4268, 1.2881], 0.2151,0 1139],(0.1 2812,0.1 468)]	[[0.4549, 1.3414], 0.1943,0 1.054],(0.1 2579,0.1 377)]	[[0.4832, 1.3459], 0.1791,0 1.024],(0.1 1894,0.0 973)]	[[0.4832, 1.3459], 0.1791,0 1.024],(0.1 1894,0.0 973)]	[[0.4804, 1.3275], 0.185,0.1 0.953],(0.1 47,0.141 3)]	[[0.4804, 1.3275], 0.185,0.1 0.953],(0.1 47,0.141 3)]	[[0.4804, 1.3275], 0.185,0.1 0.953],(0.1 47,0.141 3)]	[[0.4447, 1.3613], 0.2021,0 1.054],(0.1 2681,0.1 371)]	[[0.4422, 1.3157], 0.2021,0 1.054],(0.1 2681,0.1 371)]	[[0.4422, 1.3157], 0.2021,0 1.054],(0.1 2681,0.1 371)]	[[0.4422, 1.3157], 0.2021,0 1.054],(0.1 2681,0.1 371)]	[[0.4578, 1.3363], 0.1937,0 1.093],(0.1 2581,0.1 422)]	[[0.3687, 1.2976], 0.2477,0 1.01],(0.1 3175,0.1 437)]	
TH7	[[0.4689, 1.3259], 0.1901,0 1179],(0.1 2536,0.1 484)]	[[0.4615, 1.3482], 0.1916,0 1066],(0.1 256,0.13 85)]	[[0.4289, 1.3092], 0.2104,0 1096],(0.1 278,0.14 2)]	[[0.4347, 1.3568], 0.2145,0 1083],(0.1 2768,0.1 396)]	[[0.4829, 1.3668], 0.1762,0 1.065],(0.1 2392,0.1 372)]	[[0.4756, 1.3194], 0.1892,0 1.065],(0.1 2498,0.1 478)]	[[0.4215, 1.283], 0.1759,0 1.072],(0.1 2399,0.1 34)]	[[0.4215, 1.283], 0.1759,0 1.072],(0.1 2399,0.1 34)]	[[0.4215, 1.283], 0.1759,0 1.072],(0.1 2399,0.1 34)]	[[0.4382, 1.3816], 0.2053,0 1.081],(0.1 2707,0.1 4)]	[[0.4382, 1.3816], 0.2053,0 1.081],(0.1 2707,0.1 4)]	[[0.4382, 1.3816], 0.2053,0 1.081],(0.1 2707,0.1 4)]	[[0.4382, 1.3816], 0.2053,0 1.081],(0.1 2707,0.1 4)]	[[0.4698, 1.363], 0.1863,0 1.031],(0.2 626,0.13 49)]	[[0.3681, 1.3164], 0.243,0.1 0.97],(0.3 174,0.14 11)]	
TH8	[[0.4944, 1.301], 0.1751,0.1 125],(0.2 379,0.14 51)]	[[0.481,1 1.2776], 0.1842,0.1 084],(0.2 457,0.14 12)]	[[0.4398, 1.2776], 0.2102,0 1137],(0.1 2744,0.1 461)]	[[0.4684, 1.3218], 0.1904,0 1.032],(0.1 2519,0.1 456)]	[[0.4953, 1.3384], 0.1781,0 1.032],(0.1 2379,0.1 36)]	[[0.514,1 1.2974], 0.1603,0.1 0.81],(0.2 219,0.14 16)]	[[0.5025, 1.2897], 0.1257,0 0.641],(0.1 1829,0.0 919)]	[[0.5025, 1.2897], 0.1257,0 0.641],(0.1 1829,0.0 919)]	[[0.5025, 1.2897], 0.1257,0 0.641],(0.1 1829,0.0 919)]	[[0.4693, 1.3522], 0.198,0.1 0.15],(0.26 09,0.147 4)]	[[0.4564, 1.2987], 0.1949,0.1 0.17],(0.2 585,0.15 16)]	[[0.4675, 1.304], 0.199,0.1 0.17],(0.2 626,0.13 7)]	[[0.4675, 1.304], 0.199,0.1 0.17],(0.2 626,0.13 7)]	[[0.4861, 1.329], 0.1827,0.1 0.13],(0.2 439,0.13 7)]	[[0.3864, 1.3263], 0.245,0.1 1.08],(0.3 139,0.14 34)]	
TH9	[[0.5423, 1.3686], 0.1512,0 1026],(0.1 2127,0.1 351)]	[[0.5196, 1.3818], 0.1676,0 1025],(0.1 2278,0.1 348)]	[[0.4869, 1.3411], 0.1808,0 1066],(0.1 2437,0.1 388)]	[[0.5151, 1.397], 0.1632,0 0.973],(0.2 241,0.12 93)]	[[0.5461, 1.4032], 0.1517,0 0.996],(0.2 2103,0.1 311)]	[[0.5546, 1.363], 0.1522,0 0.996],(0.2 2115,0.1 3)]	[[0.5483, 1.4267], 0.1484,0 0.928],(0.2 2078,0.1 251)]	[[0.5483, 1.4267], 0.1484,0 0.928],(0.2 2078,0.1 251)]	[[0.5483, 1.4267], 0.1484,0 0.928],(0.2 2078,0.1 251)]	[[0.4461, 1.3514], 0.1723,0.0 0.996],(0.2 345,0.13 25)]	[[0.5006, 1.371], 0.1608,0.0 0.996],(0.2 229,0.13 26)]	[[0.5121, 1.392], 0.1673,0 0.996],(0.2 231,0.13 53)]	[[0.5115, 1.3915], 0.1609,0 1.021],(0.1 2209,0.1 337)]	[[0.4195, 1.3481], 0.2142,0 1.049],(0.1 2788,0.1 384)]	[[0.3945, 1.3963], 0.2254,0 0.969],(0.1 2923,0.1 296)]	
TH10	[[0.4945, 1.3402], 0.1795,0 1086],(0.1 2418,0.1 41)]	[[0.4738, 1.352], 0.1965,0.1 1), 4,0.1418 41)]	[[0.4517, 1.3092], 0.2006,0 1178],(0.1 2659,0.1 489)]	[[0.4668, 1.3669], 0.1957,0 1.043],(0.1 257,0.13 65)]	[[0.4979, 1.3755], 0.1795,0 1.031],(0.1 2388,0.1 354)]	[[0.5011, 1.3299], 0.1795,0 1.031],(0.1 2388,0.1 354)]	[[0.5002, 1.3551], 0.1795,0 1.062],(0.1 2393,0.1 376)]	[[0.5002, 1.3551], 0.1795,0 1.062],(0.1 2393,0.1 376)]	[[0.5002, 1.3551], 0.1795,0 1.062],(0.1 2393,0.1 376)]	[[0.4706, 1.3933], 0.1451,0.1 0.973],(0.1 655],(0.2 058,0.09 295)]	[[0.4025, 1.278], 0.1354,0 0.973],(0.1 655],(0.2 058,0.09 36)]	[[0.4848, 1.3554], 0.181,0.1 0.973],(0.1 655],(0.2 058,0.09 4)]	[[0.47,1 1.3572], 0.193,0.0 0.973],(0.1 655],(0.2 058,0.09 96)]	[[0.4868, 1.3648], 0.1854,0 1.04],(0.2 104,0.13 7)]	[[0.3898, 1.3196], 0.2361,0 0.963],(0.1 076],(0.3 32,0.139 1)]	
TH11	[[0.5524, 1.3213], 0.1474,0 1126],(0.1 208,0.14 55)]	[[0.5298, 1.3419], 0.1622,0 104],(0.2 23,0.137 2)]	[[0.5014, 1.2991], 0.1705,0 1125],(0.1 2339,0.1 45)]	[[0.5199, 1.3494], 0.1646,0 1.074],(0.1 2242,0.1 392)]	[[0.5449, 1.3618], 0.1609,0 1.02],(0.2 184,0.13 43)]	[[0.5603, 1.3202], 0.147,0.1 0.58],(0.2 2064,0.1 94)]	[[0.5594, 1.3416], 0.1603,0.1 0.973],(0.1 187,0.13 364)]	[[0.5594, 1.3416], 0.1603,0.1 0.973],(0.1 187,0.13 364)]	[[0.5594, 1.3416], 0.1603,0.1 0.973],(0.1 187,0.13 364)]	[[0.521, 1.3758], 0.1664,0 0.993],(0.2 2283,0.1 19)]	[[0.5121, 1.3246], 0.1154,0 0.965],(0.1 723,0.09 422)]	[[0.4684, 1.2731], 0.1558,0 0.965],(0.1 723,0.09 38)]	[[0.5281, 1.3557], 0.1516,0 1.016],(0.1 2194,0.1 346)]	[[0.5454, 1.3424], 0.2021,0 1.137],(0.1 2119,0.1 454)]	[[0.4332, 1.3043], 0.2021,0 1.04],(0.2 742,0.14 46)]	
TH12	[[0.4528, 1.3595], 0.1969,0 0994],(0.1 2623,0.1 328)]	[[0.4494, 1.3673], 0.195,0.1 055],(0.2 614,0.13 77)]	[[0.4119, 1.3323], 0.2208,0 1032],(0.1 2896,0.1 363)]	[[0.4156, 1.379], 0.228,0.10 49], 0.28 8)]	[[0.4731, 1.3907], 0.1752,0 0.994],(0.2 2415,0.1 318)]	[[0.4637, 1.3453], 0.1804,0.1 0.994],(0.2 537,0.13 96)]	[[0.4709, 1.369], 0.1804,0.1 0.994],(0.2 465,0.13 51)]	[[0.4709, 1.369], 0.1804,0.1 0.994],(0.2 465,0.13 51)]	[[0.4709, 1.369], 0.1804,0.1 0.994],(0.2 465,0.13 51)]	[[0.4274, 1.4039], 0.2101,0 0.996],(0.1 2768,0.1 4)]	[[0.4183, 1.3531], 0.2178,0 0.979],(0.1 2847,0.1 394)]	[[0.4513, 1.3713], 0.1909,0 0.979],(0.1 2584,0.1 307)]	[[0.3795, 1.3183], 0.1513,0 0.979],(0.1 2165,0.1 891)]	[[0.4577, 1.3795], 0.19,0.10 1), 1,0.1339 3)]	[[0.3587, 1.3403], 0.2501,0 1.063],(0.1 3235,0.1 35)]	
TH14	[[0.5322, 1.3915], 0.1671,0 0942],(0.1 2261,0.1 277)]	[[0.5212, 1.4026], 0.1697,0 0972],(0.1 2289,0.1 297)]	[[0.4928, 1.3589], 0.1759,0 1045],(0.1 2404,0.1 362)]	[[0.5161, 1.4151], 0.1665,0 0.994],(0.2 2267,0.1 271)]	[[0.5379, 1.3814], 0.1531,0 0.973],(0.2 221,0.12 86)]	[[0.5521, 1.4021], 0.1525,0 0.973],(0.2										

Table 11
NZN DEMATEL results.

Risks	$\otimes r$	$\otimes c$	$\otimes r + \otimes c$	Weight	$\otimes r - \otimes c$	Relationship	Rank
TH1	12.3593	13.1063	25.4656	0.0655	-0.747	Effect	14
TH2	12.302	13.0377	25.3397	0.06518	-0.7357	Effect	15
TH3	13.423	12.6886	26.1116	0.06716	0.7344	Cause	5
TH4	13.3709	12.9724	26.3433	0.06776	0.3985	Cause	2
TH5	12.6817	13.2686	25.9503	0.06675	-0.5869	Effect	7
TH6	12.6727	13.1657	25.8384	0.06646	-0.493	Effect	9
TH7	12.6896	13.2284	25.918	0.06666	-0.5388	Effect	8
TH8	12.7494	13.3616	26.111	0.06716	-0.6122	Effect	6
TH9	13.2644	13.0362	26.3006	0.06765	0.2282	Cause	3
TH10	12.8755	12.8446	25.7201	0.06615	0.0309	Cause	11
TH11	13.212	13.0372	26.2492	0.06752	0.1748	Cause	4
TH12	12.6505	12.9928	25.6433	0.06596	-0.3423	Effect	12
TH14	13.3234	13.129	26.4524	0.06804	0.1944	Cause	1
TH16	13.4714	12.2788	25.7502	0.06623	1.1926	Cause	10
TH17	13.3475	12.2454	25.5929	0.06583	1.1021	Cause	13

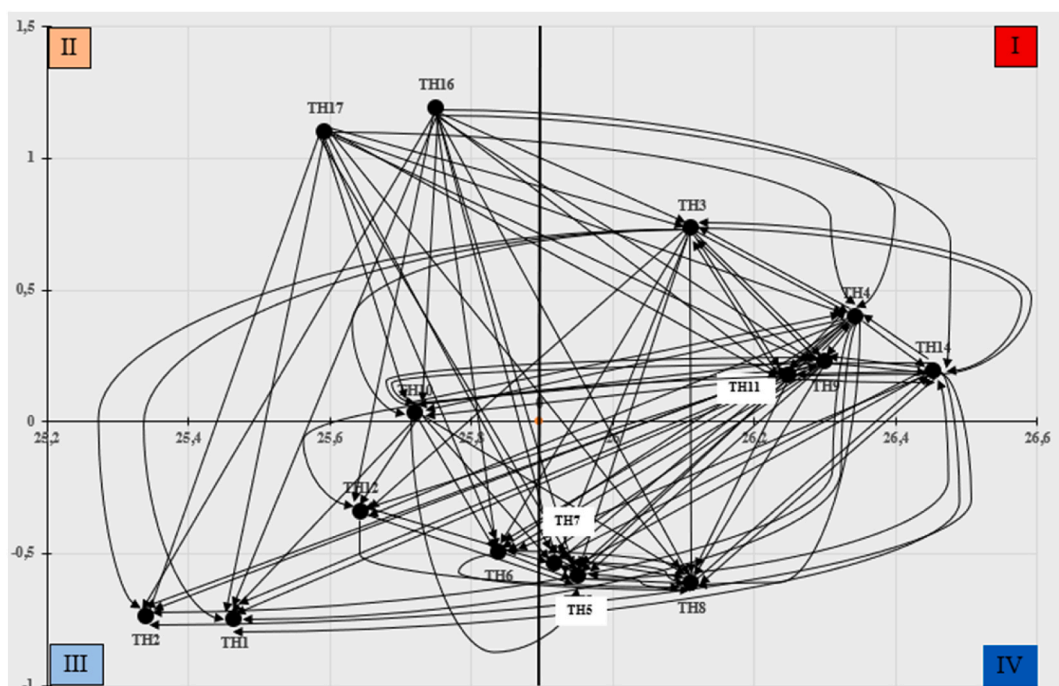


Fig. 3. Influential network relation map (INRM).

influence, including TH8, TH7, and TH5. The remaining risks, including TH6, TH12, TH1, and TH2, are categorized as effect group factors with low influence.

As outlined earlier, the five risks identified in the first quadrant exert the most significant influence. These risks are inherent to the cause group and carry substantial weight, making them pivotal factors that must be addressed to mitigate threats to the Vietnamese financial and banking system. Malware infections (TH14), for instance, serve as underlying causes contributing to various risks such as data breaches (TH7), where spyware is utilized to acquire sensitive information illicitly; ransomware attacks (TH6), wherein malware encrypts data and demands ransom for its release, and phishing attacks (TH8), where malware facilitates external actors in phishing users within internal systems and also increases cloud security risk (TH1) because malware can affect and penetrate the cloud system [13,85,86]. Moreover, malware possesses the capability to compromise authentication systems (TH11) through destructive or deceptive means and serves as a tool for man-in-the-middle attacks (TH12), thus exacerbating the threat landscape [13,86]. Malware infections emerge as fundamental causes that significantly impact a majority of the identified risks, underscoring the critical importance of addressing them to enhance the resilience and integrity of the financial system. Malware represents an external risk agent compromising system integrity, while TH11 - Weak Authentication Systems, constitutes an internal risk factor within the system. Ranked fourth in weight among the cause group, weak authentication systems significantly influence multiple risks. They heighten the system's vulnerability to penetration, particularly concerning biometric data vulnerabilities (TH10), given the advancements in AI

spoofing technology [87]. Additionally, weak authentication systems amplify threats posed by TH1, TH6, TH7, and TH8, akin to malware infections. Consequently, inadequate security and authentication systems elevate susceptibility to external threats, jeopardizing system security and integrity. TH4 - Supply Chain Vulnerabilities emerge as another significant risk factor contributing to numerous other risks. Research findings underscore its position as the second most influential risk within the cause group. Weaknesses in third-party systems render the entire infrastructure susceptible to attacks, fostering risks associated with data compromise (TH7, TH1), phishing (TH8), and even amplifying the threat of Dos attacks (TH5) due to system vulnerabilities. Inadequate security measures and authentication within third-party devices and services exacerbate risks such as TH3, TH10, and TH11, thereby increasing the likelihood of cyber attacks originating from these vulnerabilities [88]. TH9 - Insider Threats present another internal risk factor from the parties operating within the system. Unlike external threats, Insider threats entail risks from within the organization itself. Employees falling victim to phishing attacks (TH8), divulging confidential data (TH7), engaging in espionage activities, system disruption (TH12), or installing malicious software (TH14) are plausible scenarios [89]. Furthermore, insiders' inadvertent or deliberate disclosure of authentication and security-related information (TH10, 11) further heightens systemic risks. Internal threats significantly impact the banking and financial system, providing external actors with more favorable conditions to exploit vulnerabilities and undermine system integrity [13]. TH3 - Mobile Device and Web Application Security Risks rank as the fifth most weighted cause among the identified risks. Elevated Mobile Device and Web Application Security risks heighten the system's vulnerability to

Table 12
Normalized assessment matrix.

	TH1	TH2	TH3	TH4	TH5	TH6	TH7	TH8	TH9	TH10	TH11	TH12	TH14	TH16	TH17
SG1	[(0.2275, 0.2054),(0.2011,0 2067),(2095,0.2 125)]	[(0.2228, 0.2241),(0.2071,0 1836),(0 2142,0 1939)]	[(0.2735, 0.2725),(0.1464,0 1256),(0 1526),(0 1792,0.1 1454)]	[(0.2578, 0.2502),(0.161,0 1526),(0 1792,0.1 678)]	[(0.2202, 0.2183),(0.2066,0 1889),(0 2167,0 1997)]	[(0.2019, 0.2219),(0.2288,0 1833),(0 2531,0 1961)]	[(0.1833, 0.2148),(0.2502,0 1951,0 1969),(0 2537,0 2031)]	[(0.2299, 0.2508),(0.1951,0 1522),(0 1691),(0 2071,0 1671)]	[(0.2489, 0.2326),(0.1727,0 232,0.19 854)]	[(0.198,0 214),(0 2403),(0 238,0.19 36),(0.23 164),(0.2 302,0.17 8)]	[(0.2067, 0.2403),(0.221,0 2485,0 1703),(0 364),(0.2 162),(0.1 77)]	[(0.1849, 0.2338),(0.2485,0 2707,0 1451),(0 2681,0.1 252,0.17 841)]	[(0.1689, 0.256),(0 2548),(0 2126),(0 1478),(0 1447),(0 1886,0 1621)]	[(0.2147, 0.2548),(0.2126,0 1478),(0 1447),(0 1886,0 1621)]	[(0.2484, 0.2559),(0.1722,0 1447),(0 1886,0 1621)]
	SG2	[(0.244,0 2472),(0 1747,0.1 577),(0.1 93,0.170 7)]	[(0.2844, 0.2376),(0.1326,0 1663),(0 1526,0 1803)]	[(0.2281, 0.2473),(0.2006,0 1546),(0 2089,0 1707)]	[(0.2515, 0.2331),(0.1717,0 1726),(0 1855,0 1848)]	[(0.2496, 0.2331),(0.174,0 1716),(0 1874,0.1 223,0.19 849)]	[(0.214,0 2277),(0 2159,0 1789),(0 223,0.19 1688,0 1869)]	[(0.2681, 0.2311),(0.1489,0 1789),(0 1986,0 1869)]	[(0.2384, 0.2406),(0.1871,0 1698,0 1851,0.1 1773)]	[(0.2519, 0.253),(0.2408),(0.1698,0 1851,0.1 1772)]	[(0.3005, 0.2408),(0.1146,0 1615),(0 1337),(0 1523)]	[(0.2757, 0.2657),(0.1418,0 1398),(0 1612,0 1587)]	[(0.2527, 0.2593),(0.1673,0 1543),(0 1847,0 1731)]	[(0.2639, 0.2448),(0.1539,0 1593),(0 173,0.1 2044,0.1 63)]	[(0.2326, 0.255),(0 1868,0 1467),(0 2044,0.1 63)]
SG3		[(0.2632, 0.2215),(0.1556,0 1858),(0 1738,0.1 965)]	[(0.2116, 0.2486),(0.2135,0 1501),(0 2253,0 1693)]	[(0.2633, 0.2535),(0.1548,0 1468),(0 1737,0 1644)]	[(0.2882, 0.2503),(0.1284,0 1514),(0 1488,0 1677)]	[(0.2453, 0.2389),(0.1774,0 1663),(0 1917,0 1791)]	[(0.2514, 0.282),(0 2095,0 144),(0.1 855,0.13 2216,0 6)]	[(0.2153, 0.2318),(0.2614,0 156,0 1392),(0 1743,0.1 1567,0 566)]	[(0.2627, 0.2523),(0.253,0 156,0 1486),(0 1775),(0 1657)]	[(0.2803, 0.2305),(0.1594,0 157),(0 1785,0 1875)]	[(0.2763, 0.2452),(0.1387,0 157),(0 1738),(0 1622,0 1873)]	[(0.2748, 0.2307),(0.1408,0 1309),(0 1622,0 1501)]	[(0.2551, 0.2679),(0.1619,0 1309),(0 1819,0 1501)]	[(0.2565, 0.2572),(0.1619,0 1434),(0 1805,0 1607)]	[(0.2583, 0.2745),(0.1613,0 125),(0 1786,0 435)]
	SG4	[(0.21,0.2 373),(0.2 164,0.16 89),(0.22 7,0.1807)]	[(0.2322, 0.2475),(0.1913,0 1533),(0 2048,0 1704)]	[(0.2647, 0.2394),(0.1547,0 1629),(0 1623),(0 1659,0.1 764)]	[(0.2711, 0.2415),(0.146,0 1623),(0 1753),(0 2143,0 1871)]	[(0.2226, 0.2309),(0.2019,0 1747,0 1391),(0 1632),(0 1906,0 1573)]	[(0.2464, 0.2606),(0.1917,0 1747,0 1391),(0 1632),(0 1906,0 1573)]	[(0.2324, 0.2387),(0.1917,0 1521,0 1218),(0 1419),(0 1702,0 584)]	[(0.2667, 0.2762),(0.1521,0 143,0.1941 183,0.1 2235,0 212,0.2346 1826,0 1262),(0 199,0.14 63)]	[(0.2738, 0.2596),(0.2462),(0 1941,0 183,0.1 2235,0 212,0.2346 1826,0 1262),(0 199,0.14 63)]	[(0.2325, 0.2462),(0 251),(0 2425),(0 2728),(0 2658),(0 2176),(0 1826,0 1262),(0 199,0.14 63)]	[(0.2373, 0.251),(0 2425),(0 2728),(0 2658),(0 2176),(0 1826,0 1262),(0 199,0.14 63)]	[(0.2066, 0.2425),(0 2728),(0 2658),(0 2176),(0 1826,0 1262),(0 199,0.14 63)]	[(0.2146, 0.2728),(0 2658),(0 2176),(0 1826,0 1262),(0 199,0.14 63)]	[(0.1929, 0.2658),(0 2176),(0 1826,0 1262),(0 199,0.14 63)]
SG5		[(0.2218, 0.2452),(0.2039,0 1571),(0 2151,0.1 728)]	[(0.2136, 0.2554),(0.2122,0 1571),(0 2233,0.1 623)]	[(0.2442, 0.2552),(0.1784,0 1928,0.1 627)]	[(0.2377, 0.2473),(0.1832,0 1993,0 1707)]	[(0.258,0 2542),(0 1609,0 2036),(0 164	[(0.2233, 0.2392),(0.2645, 0.2494),(0.2096,0 1905,0.1345 146),(0.1 1523),(0 1706),(0 1544,0.1 2139,0 1828)]	[(0.2625, 0.2645),(0.2096,0 1905,0.1345 146),(0.1 1523),(0 1706),(0 1544,0.1 2139,0 1828)]	[(0.2158, 0.2339),(0.256,0 1905,0.1345 146),(0.1 1523),(0 1706),(0 1544,0.1 2139,0 1828)]	[(0.2334, 0.2339),(0.256,0 1905,0.1345 146),(0.1 1523),(0 1706),(0 1544,0.1 2139,0 1828)]	[(0.2826, 0.2351),(0.2015,0 188,0.1728 0.1719,0 1575),(0 167),(0.2 1685),(0 1469),(0 1304),(0 1942,0 1503)]	[(0.2348, 0.2359),(0.188,0 1728,0.1719 0.1575),(0 167),(0.2 1685),(0 1469),(0 1304),(0 1942,0 1503)]	[(0.2469, 0.2355),(0.1728,0 1575),(0 167),(0.2 1685),(0 1469),(0 1304),(0 1942,0 1503)]	[(0.2497, 0.2545),(0.1719,0 1575),(0 167),(0.2 1685),(0 1469),(0 1304),(0 1942,0 1503)]	[(0.2428, 0.2676),(0.1757,0 1575),(0 167),(0.2 1685),(0 1469),(0 1304),(0 1942,0 1503)]
	SG6	[(0.239,0 2392),(0 1824,0.1 641),(0.1 98,0.178 8)]	[(0.2205, 0.2597),(0.2044,0 1412),(0 2165,0 1583)]	[(0.25,0 2487),(0 0.2654),(0.1882,0 1516),(0.1 2015),(0 1355),(0 205,0.1 1929,0 1663)]	[(0.2354, 0.2657),(0.1882,0 1516),(0.1 2015),(0 1355),(0 205,0.1 1929,0 1663)]	[(0.2441, 0.2657),(0.2134,0 1495,0 1506),(0 1458),(0 2125,0.1 1691,0 1663)]	[(0.2116, 0.2559),(0.2342),(0.2063,0 1613),(0 1686),(0 1699),(0 2125,0.1 1691,0 1663)]	[(0.2679, 0.2559),(0.2432),(0.2063,0 1613),(0 1686),(0 1699),(0 2125,0.1 1691,0 1663)]	[(0.2156, 0.2353),(0.2362),(0.169,0 2502,0 1699),(0 1328),(0 2067,0 1818)]	[(0.2509, 0.2353),(0.2362),(0.169,0 2502,0 1699),(0 1328),(0 2067,0 1818)]	[(0.1813, 0.2362),(0.2667),(0 1945),(0 1578),(0 1522),(0 1727)]	[(0.2304, 0.2667),(0.2162),(0 1989,0 2479,0 1578),(0 1522),(0 1727)]	[(0.2249, 0.2453),(0.1989,0 2479,0 1578),(0 1522),(0 1727)]	[(0.1818, 0.2273),(0.1866,0 1662,0 1595),(0 1571),(0 1716)]	[(0.2527, 0.2446),(0.1866,0 1662,0 1595),(0 1571),(0 1716)]
SG7		[(0.2184, 0.246),(0 2097,0.1 565),(0.2 186,0.17 19)]	[(0.2489, 0.2623),(0.2441,0 2033),(0 1382),(0 188,0.1 556)]	[(0.221,0 2441),(0 2033),(0 1555),(0 216,0.17 38)]	[(0.2423, 0.2584),(0.1771,0 1711),(0.1588),(0 1941,0 589)]	[(0.2428, 0.2469),(0.1812,0 2057),(0 1427),(0 1319,0.1 1598,0 589)]	[(0.2172, 0.259),(0 2057),(0 1387,0 1884,0 1599),(0 2038,0 2108,0 1742)]	[(0.2772, 0.2689),(0.2323),(0.1884,0 1986,0 1599),(0 2038,0 2108,0 1742)]	[(0.2331, 0.2323),(0.1884,0 1986,0 1599),(0 2038,0 2108,0 1742)]	[(0.2262, 0.2437),(0.2362),(0.1724,0 1578,0 1522),(0 1727)]	[(0.2467, 0.2362),(0.2746),(0 1578,0 1522),(0 1727)]	[(0.2614, 0.2746),(0.1578,0 1522),(0 1727)]	[(0.2582, 0.249),(0 1596,0 2181),(0 161),(0.2 1411),(0 1487),(0 2216,0.1 649)]	[(0.2154, 0.2413),(0.2595),(0 2115,0 2283),(0 1487),(0 2216,0.1 649)]	[(0.1971, 0.2595),(2115,0 2283),(0 1487),(0 2216,0.1 649)]
	SG8	[(0.2142, 0.2749),(0.2094,0 123),(0.2 228,0.14 3)]	[(0.23,0 2408),(0 1925,0.1 629),(0.2 0,0.177 2)]	[(0.2496, 0.241),(0 1719,0 2035,0 1613),(0 1874,0.1 769)]	[(0.2215, 0.217),(0 0.2334),(0.1897,0 1699),(0 2107,0 1845)]	[(0.2263, 0.2086),(0.263),(0 0.208,0 1776,0 2002),(0 1377),(0 2147,0 549)]	[(0.2205, 0.2086),(0.263),(0 0.208,0 1776,0 2002),(0 1377),(0 2147,0 549)]	[(0.2416, 0.2438),(0.2035,0 1776,0 2002),(0 1377),(0 2147,0 549)]	[(0.2223, 0.2438),(0.2035,0 1776,0 2002),(0 1377),(0 2147,0 549)]	[(0.2486, 0.2437),(0.2523),(0.2499),(0 1611),(0 1499),(0 1883,0 2196,0 1743)]	[(0.2173, 0.2523),(0.2499),(0 1611),(0 1499),(0 1883,0 2196,0 1743)]	[(0.2638, 0.2507),(0.1546,0 175,0.19 3,0.1672 1681)]	[(0.2444, 0.2507),(0.1546,0 175,0.19 3,0.1672 1681)]	[(0.23,0 245),(0.1 955,0.15 0.192,0 20.192 1748)]	[(0.2281, 0.2431),(0.1969,0 1885,0 1478)]
SG9		[(0.225,0 255),(0.1 99,0.14 6),(0.212, 0.1629)]	[(0.2613, 0.2558),(0.1577,0 1457),(0 1757,0 1622)]	[(0.2379, 0.2703),(0.1839,0 1293),(0 1476)]	[(0.2862, 0.2588),(0.1295,0 1421),(0 1592)]	[(0.2297, 0.2737),(0.1925,0 1253),(0 2073,0 1443)]	[(0.2331, 0.2499),(0.271),(0 0.1898,0 2035,0 168)]	[(0.2191, 0.271),(0 0.1639,0 2179,0.1 47)]	[(0.2549, 0.2421),(0.2621),(0.1606,0 164),(0.1 1821,0.1 1797,0 758)]	[(0.2572, 0.2621),(0.2369),(0.2272,0 0.207,0 1383),(0 168),(0 2324,0.1 2184,0.1 511)]	[(0.2046, 0.2369),(0.2653),(0.207,0 0.2055,0 1638),(0 1342),(0 2184,0.1 526)]	[(0.2186, 0.2384),(0.2653),(0.207,0 0.2055,0 1638),(0 1342),(0 2184,0.1 526)]	[(0.2207, 0.2384),(0.2653),(0.207,0 0.2055,0 1638),(0 1342),(0 2184,0.1 526)]	[(0.2285, 0.2657),(0.1921,0 1586,0 1355),(0 1262),(0 1912,0 1663)]	[(0.2586, 0.2716),(0.1586,0 1496),(0 1912,0 1663)]
	SG10	[(0.2774, 0.279),(0 1402,0.1 196),(0.1 596,0.13 9)]	[(0.2398, 0.2601),(0.182,0 14),(0.19 72,0.157 9)]	[(0.2504, 0.2531),(0.1719,0 1485),(0 1866,0 1649)]	[(0.2602, 0.2615),(0.1575,0 1371),(0 1768,0 1564)]	[(0.2365, 0.2427),(0.1871,0 1604),(0 2005,0 1752)]	[(0.207,0 2436),(0.2212,0 1605),(0 23,0.174 3)]	[(0.2719, 0.2563),(0.1441,0 1432),(0 2168,0 1616)]	[(0.2202, 0.2563),(0.2032,0 1432),(0 2168,0 1616)]	[(0.275,0 2614),(0.2438),(0.1936,0 1391),(0 1572,0.1 2018,0 1742)]	[(0.2351, 0.2438),(0.1936,0 1391),(0 1572,0.1 2018,0 1742)]	[(0.2797, 0.2634),(0.139,0 1391),(0 1572,0.1 2018,0 1742)]	[(0.2568, 0.2589),(0.1627,0 1391),(0 1572,0.1 2018,0 1742)]	[(0.2317, 0.2496),(0.196,0 1391),(0 1572,0.1 2018,0 1742)]	[(0.27,0 2496),(0 1467,0.1 515),(0.1 67,0.168 4)]
Weight		0.0655	0.06518	0.06716	0.06776	0.06675	0.06646	0.06666	0.06716	0.06765	0.06615	0.06752	0.06596	0.06804	0.06623

infiltration from these devices, exposing it to a range of risks outlined by TH1, TH5, TH6, TH7, TH8, and TH9. Weaknesses in system structure, alongside inadequate authentication systems, contribute to the prevalence of attacks targeting mobile devices and websites [90]. This common risk serves as a precursor to numerous other risks listed, emphasizing its critical role in the system's security landscape. As highlighted earlier, TH16 and TH17 exhibit the largest net influence value despite falling within quadrant II. According to Javaheri [13], IoT Vulnerabilities catalyze risks such as DOS attacks (TH5) and susceptibility to malware (TH14), leading to system vulnerabilities. Cyber attackers can exploit IoT vulnerabilities to compromise data security (TH7) or disrupt cloud systems (TH1). Moreover, heightened IoT risks exacerbate the likelihood of cyber attacks, impacting the overall system safety and integrity (TH5, 6, 8, 12) [91]. Given its wide-reaching implications, proactive measures against IoT vulnerabilities are imperative. TH17 - Insecure Application Programming Interfaces (APIs) are a significant net impact value risk. Vulnerabilities in APIs can be exploited to bypass authentication mechanisms or inject malicious code into the system (TH14), facilitating unauthorized access and fraudulent activities. Similar to IoT vulnerabilities, API-related risks are connectivity-related, increasing the likelihood of cyber attacks such as TH5, 6, 8, and 12 while also heightening risks associated with data security and cloud systems (TH1, 7) [63]. These vulnerabilities underscore the importance of addressing API security to mitigate broader cybersecurity threats. As elucidated above, risks within quadrants III and IV, attributed to the effect group, predominantly arise or escalate in likelihood due to the influence of risks originating from the cause group. TH8, TH5, TH7, TH9, TH6, TH1, TH2, and TH12 represent common cyber attack vectors orchestrated by external actors. These risks become more pronounced when the system experiences vulnerabilities from internal disruptions, weak authentication processes, or malware-induced compromises to security and system integrity. Factors within this effect group significantly influence the financial and banking system, exemplified by TH8, TH7, and TH5, ranked sixth, seventh, and eighth in impact weights, respectively. All 15 risks above carry notable implications for the Vietnamese financial and banking system, underscoring the imperative to devise comprehensive solutions to mitigate these risks. Priority should be accorded to addressing causes within the cause group, given their propensity to catalyze and exacerbate numerous other risks across the system.

4.5. NZN COCOSO results

Experts will assess the ten proposed strategies to determine their effectiveness in minimizing and mitigating risks. Experts will evaluate these strategies using the scale provided in Table 6. The combined evaluation results from 30 experts, standardized according to Equation (28) and (29), are presented in Table 12:

The data in Table 13 is processed using Equation (30)–(41) to calculate the results Λ_i , Ψ_i , and C_{ii} , C_{iii} , C_{iiii} , along with the ranking order of the strategies. The results are presented in Table 13 below:

The ranking results of 10 strategies using the NZN COCOSO method are as follows: SG10>SG3>SG9>SG2>SG4>SG5>SG7>SG6>SG8>SG1. Thus, SG10, SG3, and SG9 are the top 3 strategies rated best in facing Vietnam's financial and banking system risks. SG10 - Investing in Advanced Threat Detection Systems is crucial for addressing essential banking and financial sector cybersecurity risks. Enhancing the system contributes significantly to the early detection of security vulnerabilities and system flaws, enabling prompt repairs and remediation actions [92,93]. This proactive approach reduces the susceptibility to various types of cyber attacks, mitigating the risk of system infiltration and thereby diminishing risks such as TH3, TH5, TH6, TH7, TH8, TH12, TH16, and TH17. Upgrading threat detection systems enables swift identification of malicious codes and programs (TH14) attempting to breach the system, as well as vulnerabilities in security protocols (TH10, 11), and errors stemming from hardware, connections, or third-party involvements (TH4, TH16) [94]. A proactive defensive strategy entails early risk detection and mitigation, offering a comprehensive approach to addressing the abovementioned spectrum of risks [73]. Financial institutions strategically deploying such solutions fortify their defenses against emerging threats, enhancing overall resilience to cybersecurity challenges. SG3 - Implementing Robust Employee Training Programs emerges as the second-highest-rated strategy among the top three. This strategy focuses on enhancing employee competencies and cultivating a vigilant stance toward mitigating cybersecurity risks [66]. Its primary objective is to minimize insider threats by fostering employee awareness of cybersecurity indicators, thereby reducing both inadvertent incidents and intentional violations. By bolstering employee awareness, the likelihood of falling victim to external cyber-attacks such as Phishing (TH8) or Malware (TH14) diminishes as employees become more discerning with emails and

Table 13
NZN COCOSO results.

Strategies	Λ_i	Ψ_i	$\Lambda_i + \Psi_i$	C_{ii}	C_{iii}	C_{iiii}	C_i	Rank
SG1	0.6596	0.6588	0.7198	0.099	0.3333	0.9815	0.79002	10
SG2	0.6686	0.6683	0.7299	0.1004	0.4202	0.9952	0.85283	4
SG3	0.6702	0.6697	0.7315	0.1006	0.4363	0.9975	0.86392	2
SG4	0.6665	0.666	0.7272	0.1	0.4064	0.992	0.84236	5
SG5	0.6662	0.6659	0.7271	0.1	0.4020	0.9917	0.83951	6
SG6	0.6643	0.6639	0.7248	0.0996	0.3866	0.9888	0.82808	8
SG7	0.666	0.6656	0.7267	0.0999	0.4025	0.9913	0.8395	7
SG8	0.6643	0.6639	0.7251	0.0997	0.3831	0.9888	0.82604	9
SG9	0.6681	0.6678	0.729	0.1002	0.4237	0.9945	0.85435	3
SG10	0.671	0.6705	0.7324	0.1007	0.4454	0.9987	0.87008	1
Max	0.6718	0.6715	0.7324			$\varpi = 0.5$		
Min	0.6596	0.6588	0.7198					

suspicious links while also promptly reporting any anomalies to IT personnel [95]. Risks associated with TH1, TH5, TH6, TH11, TH12, TH16, and TH17 are likewise mitigated as employees augment their skills and assume greater responsibility in averting cybersecurity threats. Implementing robust employee training programs not only fortifies the organization against internal vulnerabilities but also creates a culture of cybersecurity consciousness that permeates throughout the institution, thus enhancing overall resilience to cybersecurity risks [96]. SG9 - Enforcing Multi-Factor Authentication (MFA) is recognized as the third-best strategy among the ten recommended approaches. This strategy directly addresses the risk posed by TH11 - Weak Authentication Systems. Implementing MFA fortifies the authentication system with advanced and precise authentication measures, thereby mitigating TH10 - Biometric Data Vulnerabilities risk and enhancing detection and prevention capabilities (Kebande et al., 2021 [97]). Adopting MFA enhances the protection of websites and applications (TH3) and reduces the risk of intrusion resulting from third-party errors (TH4). Consequently, the overall risk stemming from external attacks is diminished, lowering the likelihood of data breaches and cloud security vulnerabilities (TH1, TH7) [87]. This proactive solution strengthens system defenses, upgrades authentication protocols, and minimizes unauthorized access from external entities [72]. Therefore, these three strategies have effectively addressed and mitigated most of the risks outlined earlier. Primarily defensive, they involve upgrading defense systems to proactively detect threats early on while fortifying and minimizing internal threats within the organization.

4.6. Sensitive analysis

To assess the robustness of the ranking results obtained through the NZN COCOSO method, a sensitivity analysis was performed by varying the ϖ value from 0.1 to 0.9. This procedure serves to mitigate potential human judgmental biases that could influence decision outcomes [98]. Table 14 below presents the value of $\mathbb{C}_i$. Under different ϖ values.

The ranking order of strategies based on the $\mathbb{C}_i$ value is displayed in Fig. 4 below. Across varying values of ϖ from 0.1 to 0.8, the ranking order of the ten strategies remains consistent. However, with ϖ set at 0.9, the ranking order of SG5 and SG7 shifts, while the positions of the remaining eight strategies remain unchanged. Notably, the top 5 ranking order remains consistent across different ϖ values, with SG10 consistently rated as the highest-ranking strategy. This underscores the reliability and certainty of the ranking results.

4.7. Comparative analysis

A comparative analysis assessed the accuracy and reliability of the ranking results obtained through the NZN COCOSO method [99]. This analysis involved comparing the rankings generated by two other methods: NZN TOPSIS and NZN MOORA. Fig. 5 below presents the results of ranking the ten strategies using all three methods, allowing for a comprehensive comparison of their outcomes.

Fig. 4 illustrates the ranking outcomes from the three methods, consistently positioning strategies SG10 at the top and SG3 in the third position. Notably, SG10, SG3, SG2, and SG9 consistently rank among the top four strategies across all three methods. Table 15 below presents the results of the correlation analysis conducted among the ranking orders derived from the three methods, providing further insights into the alignment and consistency of the rankings.

The results of the Pearson correlation analysis indicate that the correlation coefficient between pairs of ratings exceeds 0.6, and the Sig value is less than 0.05, as indicated in Table 15. These findings suggest a strong positive correlation among the sets of rating results [100]. Combined with the earlier comments regarding the ranking orders, it becomes evident that the ranking results derived from the NZN COCOSO method are reliable and consistent.

5. Conclusion

Theoretically, this study integrates NS theory with Z-number theory to enhance the reliability and accuracy of data analysis while capturing the level of certainty among respondents. By employing this innovative approach, the data reflects higher accuracy, leading to more precise results. The proposed method combines NZN and Z-number theories with MCDM methods such as DELPHI, COCOSO, TOPSIS, and MULTIMOORA to elevate the accuracy of data analysis. In practical terms, this research reveals 15 cybersecurity risks impacting Vietnam's financial and banking system, with Malware Infections and Supply Chain Vulnerabilities emerging as the most

Table 14
 $\mathbb{C}_i$ under different ϖ values.

ϖ		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
$\mathbb{C}_i$	SG1	0.78989	0.78989	0.78993	0.78998	0.79002	0.79002	0.79006	0.79011	0.79015
	SG2	0.85283	0.85283	0.85283	0.85283	0.85283	0.85283	0.85283	0.85283	0.85283
	SG3	0.86383	0.86387	0.86387	0.86387	0.86392	0.86392	0.86392	0.86396	0.86396
	SG4	0.84227	0.84231	0.84231	0.84231	0.84236	0.84236	0.84236	0.8424	0.8424
	SG5	0.83951	0.83951	0.83951	0.83951	0.83951	0.83951	0.83951	0.83951	0.83951
	SG6	0.82804	0.82804	0.82804	0.82804	0.82808	0.82808	0.82808	0.82808	0.82808
	SG7	0.83945	0.83945	0.8395	0.8395	0.8395	0.8395	0.8395	0.8395	0.83954
	SG8	0.826	0.826	0.826	0.826	0.82604	0.82604	0.82604	0.82604	0.82604
	SG9	0.85435	0.85435	0.85435	0.85435	0.85435	0.85435	0.85435	0.85435	0.85435
	SG10	0.86999	0.87003	0.87003	0.87003	0.87008	0.87008	0.87008	0.87008	0.87012

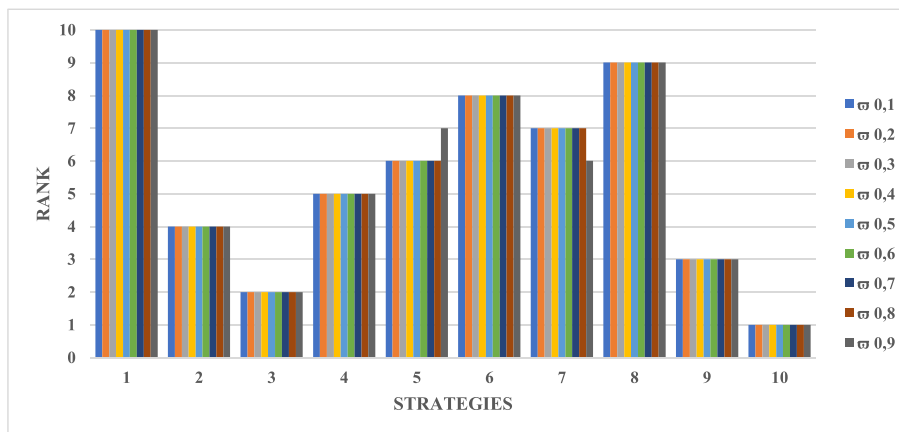


Fig. 4. Sensitive analysis results.

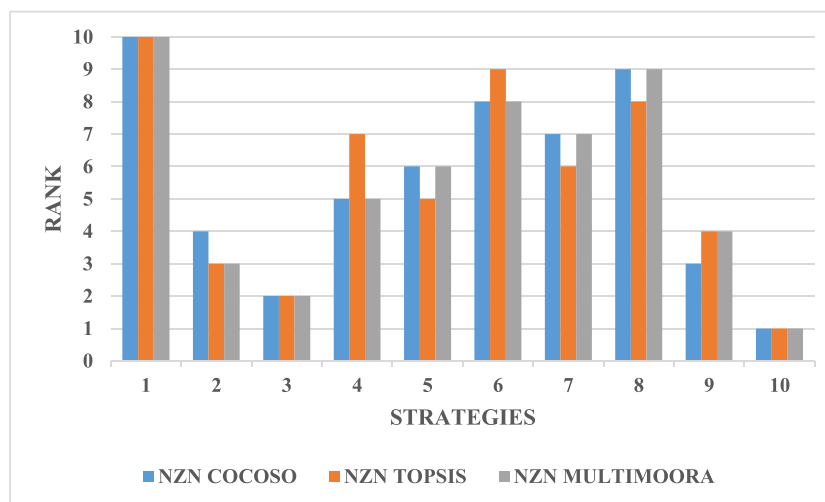


Fig. 5. Comparative results.

influential risks. These two risks serve as two of the seven root cause factors significantly affecting other risks and the overall security and integrity of Vietnam's banking system. Investing in Advanced Threat Detection Systems is the highest-rated strategy for mitigating these risks. This strategy is particularly beneficial as it directly addresses the root causes of many cybersecurity threats, enhancing the overall security and resilience of Vietnam's financial and banking infrastructure. Implementing advanced threat detection systems can prevent malware infections and mitigate supply chain vulnerabilities, ensuring a more secure financial environment. The proposed strategies offer significant benefits for Vietnam's financial and banking system. Advanced threat detection systems enable proactive identification and neutralization of threats, safeguarding sensitive data and building consumer and stakeholder trust. By addressing root causes, these strategies enhance the resilience and adaptability of the cybersecurity framework. The implications for Vietnam are substantial. Strengthened cybersecurity will protect financial transactions and data, support economic stability, and maintain investor confidence. Furthermore, these strategies can serve as a model for other developing countries facing similar cybersecurity challenges,

Table 15

Pearson correlation analysis.

		NZN COCOSO	NZN TOPSIS	NZN MULTIMOORA
NZN COCOSO	Pearson Correlation	1	0.939 ^a	0.988 ^a
	Sig. (2-tailed)		0	0
NZN TOPSIS	Pearson Correlation	0.939 ^a	1	0.952 ^a
	Sig. (2-tailed)	0		0
NZN MULTIMOORA	Pearson Correlation	0.988 ^a	0.952 ^a	1
	Sig. (2-tailed)	0	0	

^a Correlation is significant at the 0.01 level (2-tailed).

promoting a more secure and stable financial environment globally. This study demonstrates the value of integrating advanced analytical methods to identify and manage cybersecurity risks, providing actionable strategies to safeguard Vietnam's financial and banking system and support its economic growth.

However, the study has some limitations. Although the sample size exceeds the standard of 10–14 responses, increasing it further would enhance the study's accuracy. Additionally, the scope is limited to Vietnam, making the findings not globally applicable. In the comparative analysis, only TOPSIS and MULTIMOORA were used, omitting other ranking methods. Future research should integrate the NZN model with other ranking methods such as VIKOR, EDAS, COPRAS, and other MCDM. Expanding the scope beyond a single country and improving the quality of the expert panel will also be objectives to increase the study's accuracy and reliability.

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Data availability

Data will be made available on request.

CRediT authorship contribution statement

Phi-Hung Nguyen: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **The-Vu Pham:** Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Lan-Anh Thi Nguyen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis. **Hong-Anh Thi Pham:** Writing – original draft, Visualization, Validation, Resources, Formal analysis. **Thu-Hoai Thi Nguyen:** Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Tra-Giang Vu:** Writing – original draft, Visualization, Validation, Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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